Swing 3

User Manual · Version 3.0

EON Studios

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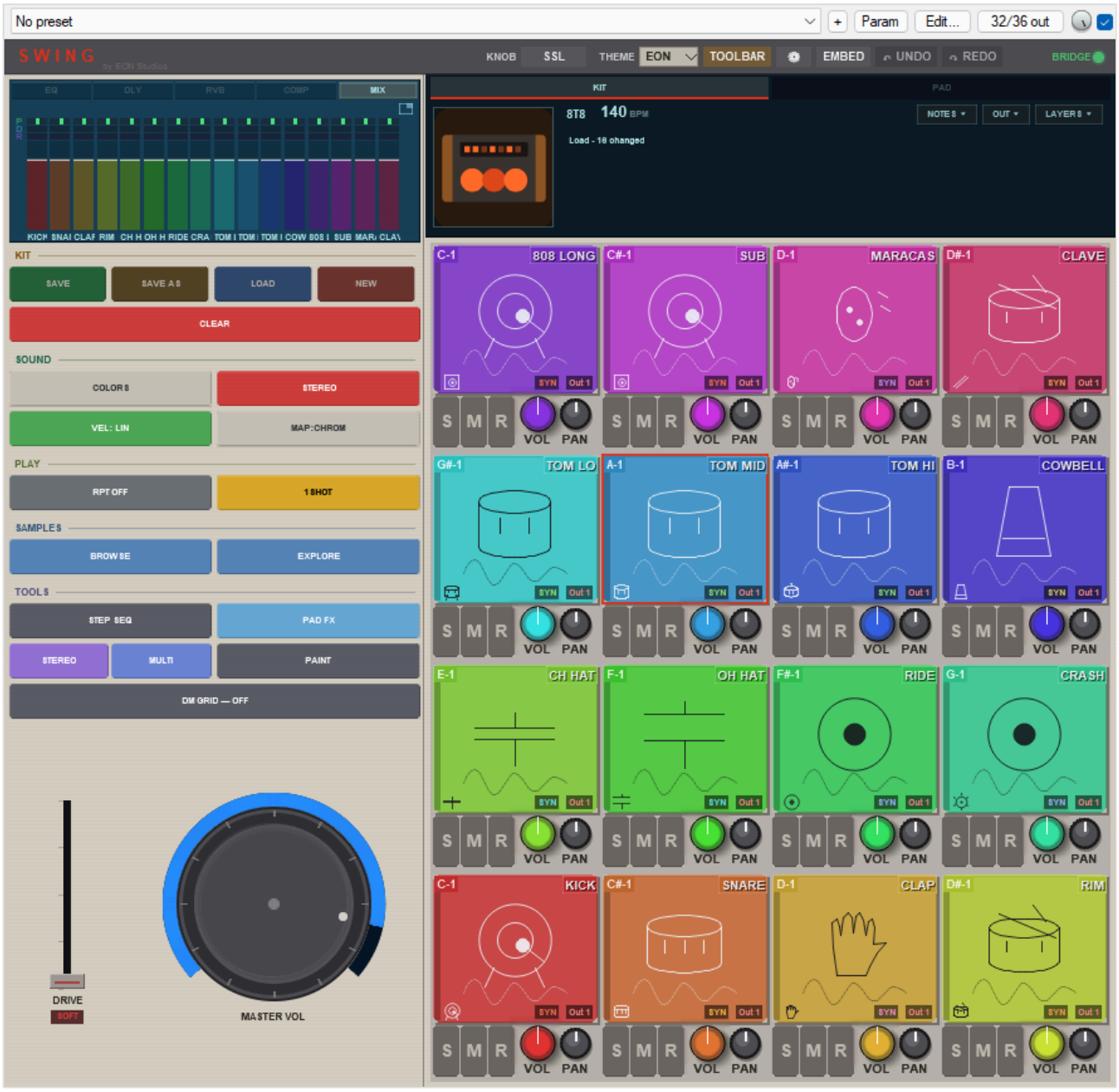
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1. Overview

Swing is a 16-pad drum sampler instrument built as a JSFX plugin for REAPER. Each pad can play a sample (or up to four layers), shaped by an ADSR envelope and routed to any of 16 stereo output buses. Every pad has its own effects chain — filters, EQ, saturation, bitcrush and delay/reverb sends — and the left-side LCD tabs expose master EQ, delay, reverb, compressor and mixer controls.



What's new in v3. Three changes are big enough to alter how you work:

- **Every pad can be a synth.** Nine engine families, thirty voice types, no samples required. See chapter 7.
- **The pad grid doubles as a workspace.** Ten full-window views open over it — note mapping, colours, routing, the kit library, the synth editor, the mixer and the four master FX pages. See chapters 9 and 10.
- **Every instance is its own instrument.** Run as many Swings as you like with no cross-talk. See chapter 18.

Swing integrates with two background helpers — the **Swing Kit Bridge** (a Lua service handling kit save/load, batch import, chop-to-pads and relinking) and the **Swing Browser** (an ImGui sample browser). Version 3 adds four companion plugins: the **Step Sequencer**, the **Drum Matrix**, the **Drum Strip** (with its **FX Return View**) and **EON Lens**.

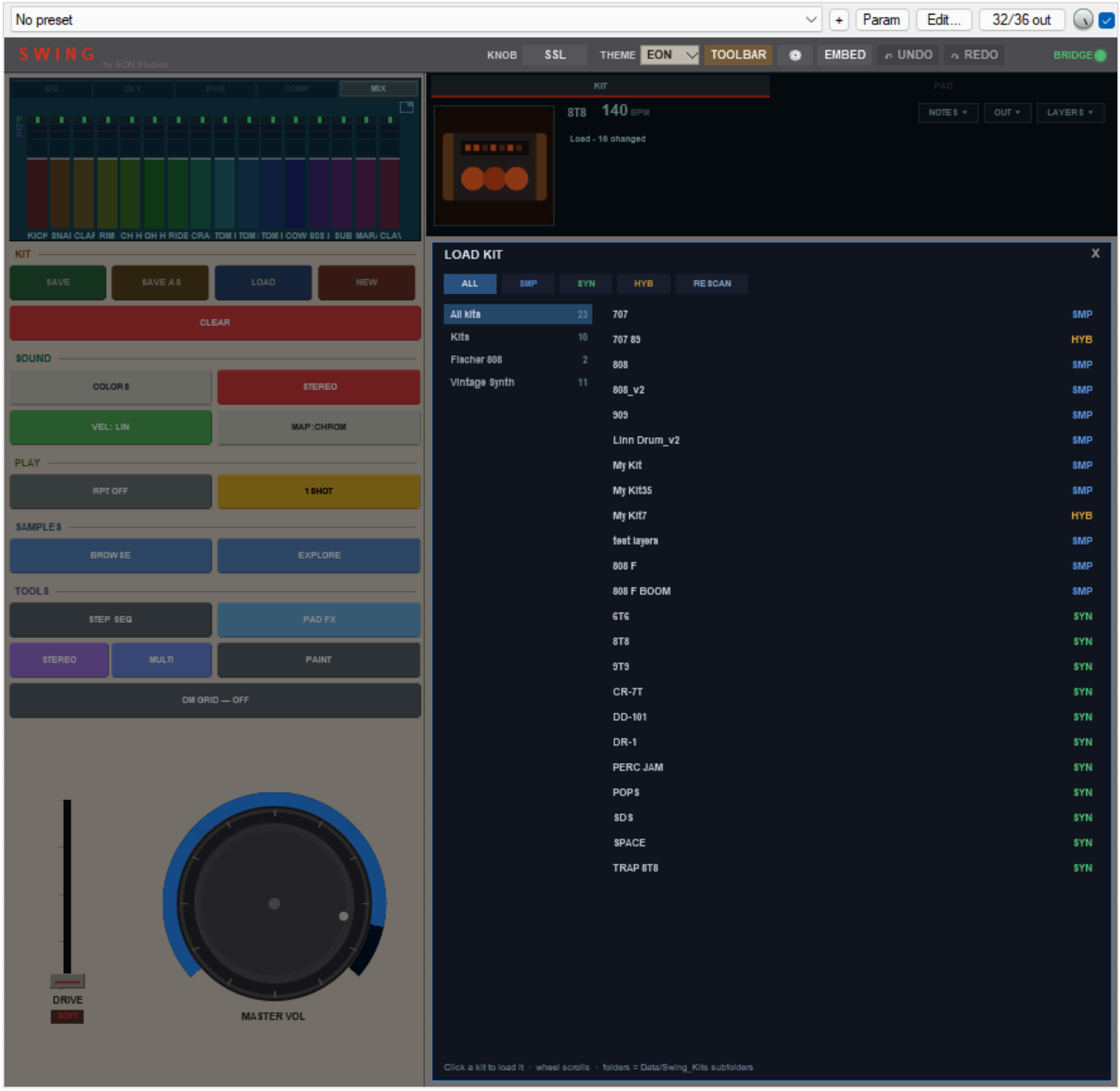
Requirements. REAPER 7.0 or later, plus three ReaPack extensions: **ReaImGui** (the sample browser), **SWS** (sample preview in the browser, and a few sync actions) and **js_ReaScriptAPI** (the Drum Matrix). Swing's JSFX itself needs none of them.

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2. Quickstart

Load Swing on a track. Insert `Swing_ReaKit` from the FX browser (`EON` → `Swing`). It opens on the last kit you used, or a factory kit on a fresh install.

Load a kit. Click **LOAD**. This opens the Kits view, where the factory library and your own kits are browsable by folder.



Load your own samples. Drag audio files onto pads, from the Swing Browser or REAPER's Media Explorer. Multi-file drags fill consecutive pads. Pads auto-name and auto-categorise by drum type where the filename allows it.

Play from MIDI. Pads map to a root note (default C2 = MIDI note 36) plus pad index. Record or draw MIDI notes in the MIDI editor, or click a pad to audition it.

Save your kit. SAVE overwrites; **SAVE AS** prompts for a name. A `.swing` file stores pad assignments, tuning, levels, colours, routing, envelopes, FX settings, synth voices and the cover image.

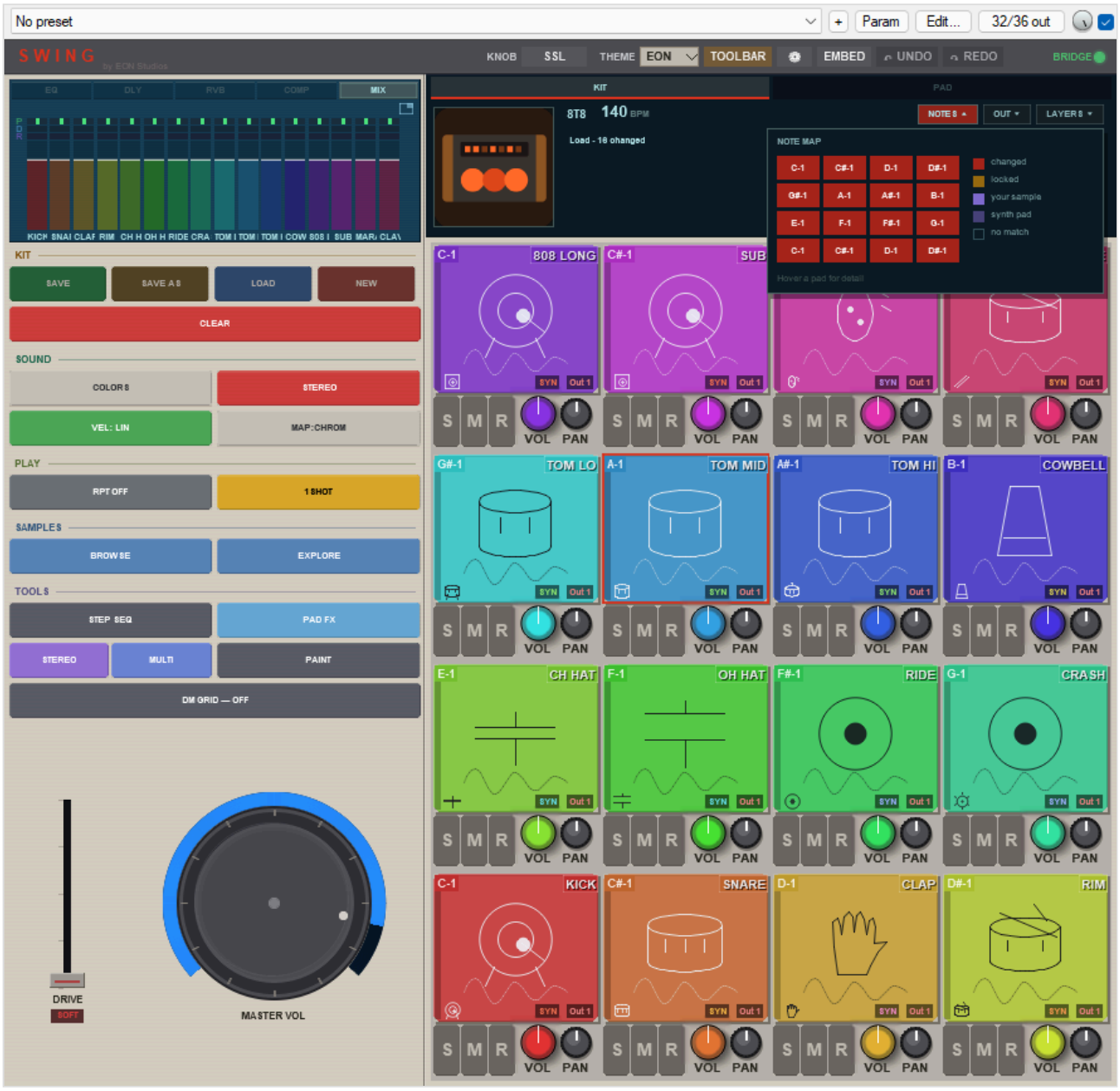
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3. User Interface Tour

Pad grid. Sixteen pads show colour, name, category glyph, note, output bus and a waveform thumbnail. Below each pad is a control strip: `S` solo, `M` mute, `R` repeat, and VOL and PAN knobs. Click a pad to trigger it; right-click for pad actions.

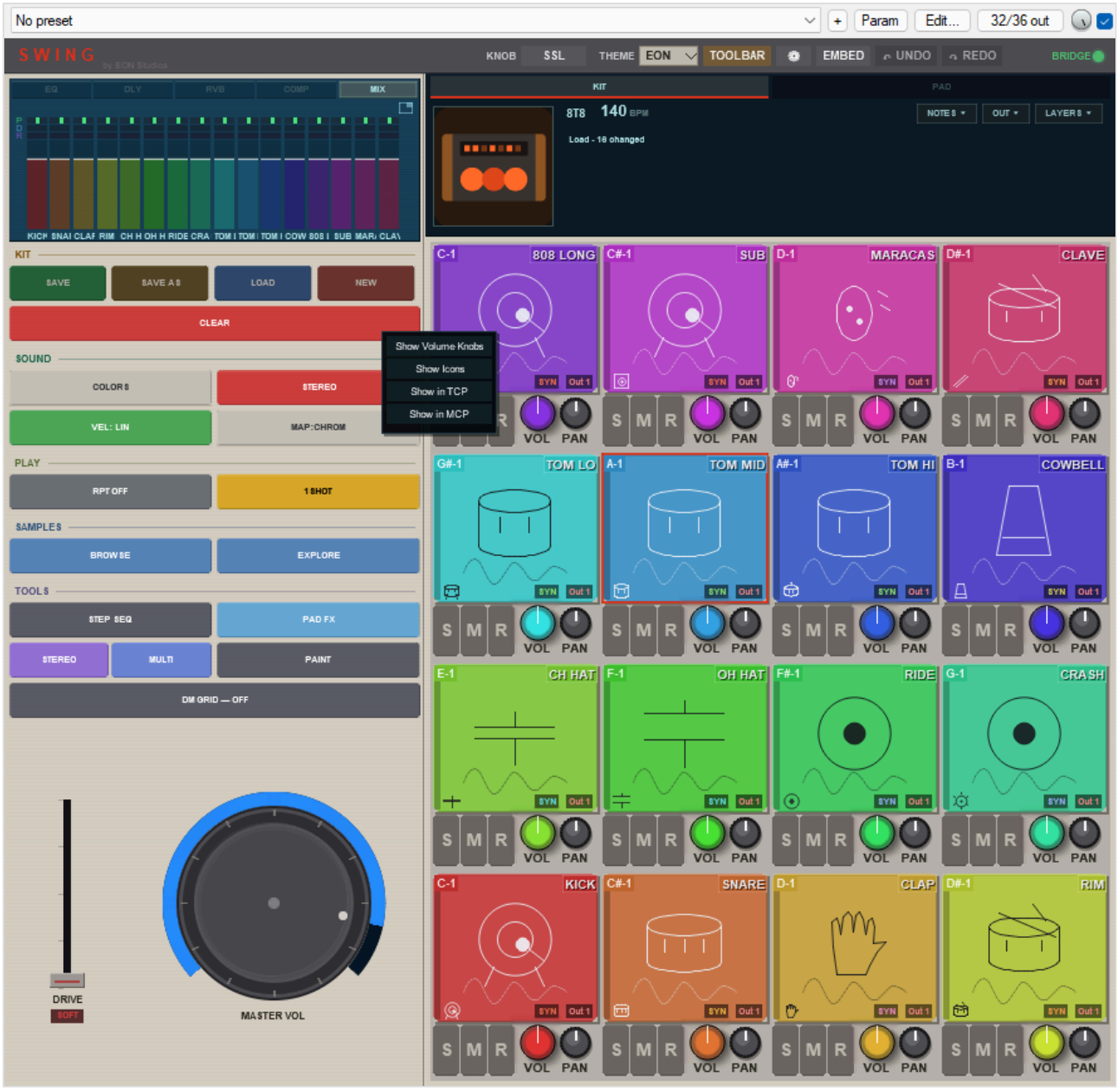
Left panel. Kit controls (SAVE, SAVE AS, LOAD, NEW, CLEAR), sound options, the tools row, and master volume and drive.

Right panel. Switches between **KIT** and **PAD** tabs. The KIT tab shows the cover image, project tempo and three dropdown panels — NOTES, OUT and LAYERS.



LCD tabs. Along the top left — EQ, DLY, RVB, COMP, MIX. These are the master processing pages and apply globally. See chapter 11.

Embedded mode. Shown in REAPER's track panel (TCP) or mixer (MCP), Swing renders a compact pad grid for quick access. The gear menu chooses what it shows and where — volume knobs, category icons, TCP, MCP:



The leftmost channel in the mixer figure in chapter 15 shows the result — Swing's compact grid living in the MCP beside its own drum channels.

4. Pads & Playback

Gain / Pan / Tune. Per-pad knobs set level, stereo position and pitch in semitones. In Repitch mode the tune knob responds immediately. Keep an eye on master output to avoid clipping.

ADSR. Attack shapes the transient, decay and sustain shape the body, release controls the tail after note-off. For one-shots, keep release moderate to avoid clicks.

Offset. Moves the playback start point into the sample. This is the single most useful control for tightening a kick: push past the silence at the front and the hit lands where you put it.

Reverse. Plays the pad backwards, for transitions and risers.

One-shot. Ignores note-off and plays to completion — right for drum hits, wrong for anything you want to cut short.

Choke groups. Assign pads to a group (1–8) so triggering one stops another. The classic use is an open and closed hi-hat sharing a group.

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5. Pad Banks

New since 3.0.18.

Swing holds four banks of sixteen pads, A to D, all live to MIDI at the same time, one on screen at a time. Think of an MPC program: drums in A, 808s and melodic pads in B, chops in C, D spare. Every kit you already have is a one-bank kit and loads into bank A unchanged.

The bank strip. Four tabs sit above the pad grid: the letter, a dot when the bank holds sound, and the bank's kit name ("kit" until a load names it, "empty" while it has none). Click a tab to switch the view. Nothing stops: a bank switch is a view, never a voice event, so a sound ringing in A keeps ringing while you look at B. Everything pad-scoped follows the tab — pad detail, layers, notes, per-pad FX, the console's pad strips, mapping, colours, outputs. Buses, sends and the master section are shared and stay where they are. The free-time readout at the strip's right end is the instrument's; the kit bar shows what the bank on screen holds, "1:12 in B".

Notes. Each bank takes the next block of sixteen notes, so all sixty-four pads are playable from one keyboard. A pad with a locked note keeps it in any bank. The Mapping view frames the bank on screen on the keyboard, with another bank's sounding pad shown dimmed in its own colour. The MIDI editor's note names list every bank's pads at their notes, all four banks at once; they follow a kit or bank load, a rename or a note change, never the bank tab.

Kits and banks. In the Kits view, **LOAD** replaces the whole instrument, all four banks. Right-click a kit for **Load into bank A / B / C / D**, or drag it onto a bank tab, to load it into that bank alone. Right-click a bank tab for its own menu: **Load kit into bank…**, **Save bank as kit…**, **Clear bank** and **Clear all four banks**. **SAVE** writes the whole instrument. A kit saved from bank A alone is the same file a 3.0 Swing wrote, so Lite and older Swings still open it; a four-bank kit opened by an older Swing shows its bank A.

Keys and controllers. Six actions join the catalogue — *EON: Swing bank A*, *B*, *C*, *D*, *next bank* and *previous bank* — bindable to keys, the toolbar and controllers like the Big View actions.

Memory. A bank has no quota of its own; only the instrument can be full. The MEM readout counts in seconds at each pad's stored rate, and a load that does not fit is refused — the banner says *Instrument full*, names the pad and tells you to clear one; nothing is trimmed. A layer now holds as much as a pad does.

Tuned and Stretch on any bank. The pitch engine has sixteen spots, shared by all four banks: any sixteen pads can be Tuned or Stretch at once, wherever they sit. A seventeenth waits — the LCD says *Waiting: all 16 Tuned / Stretch spots in use* — and takes a spot as soon as one is freed.

Dropping samples. A sample dragged from the browser lands on the pad you drop it on, in the bank you are looking at. Chop and batch import fill the bank on screen.

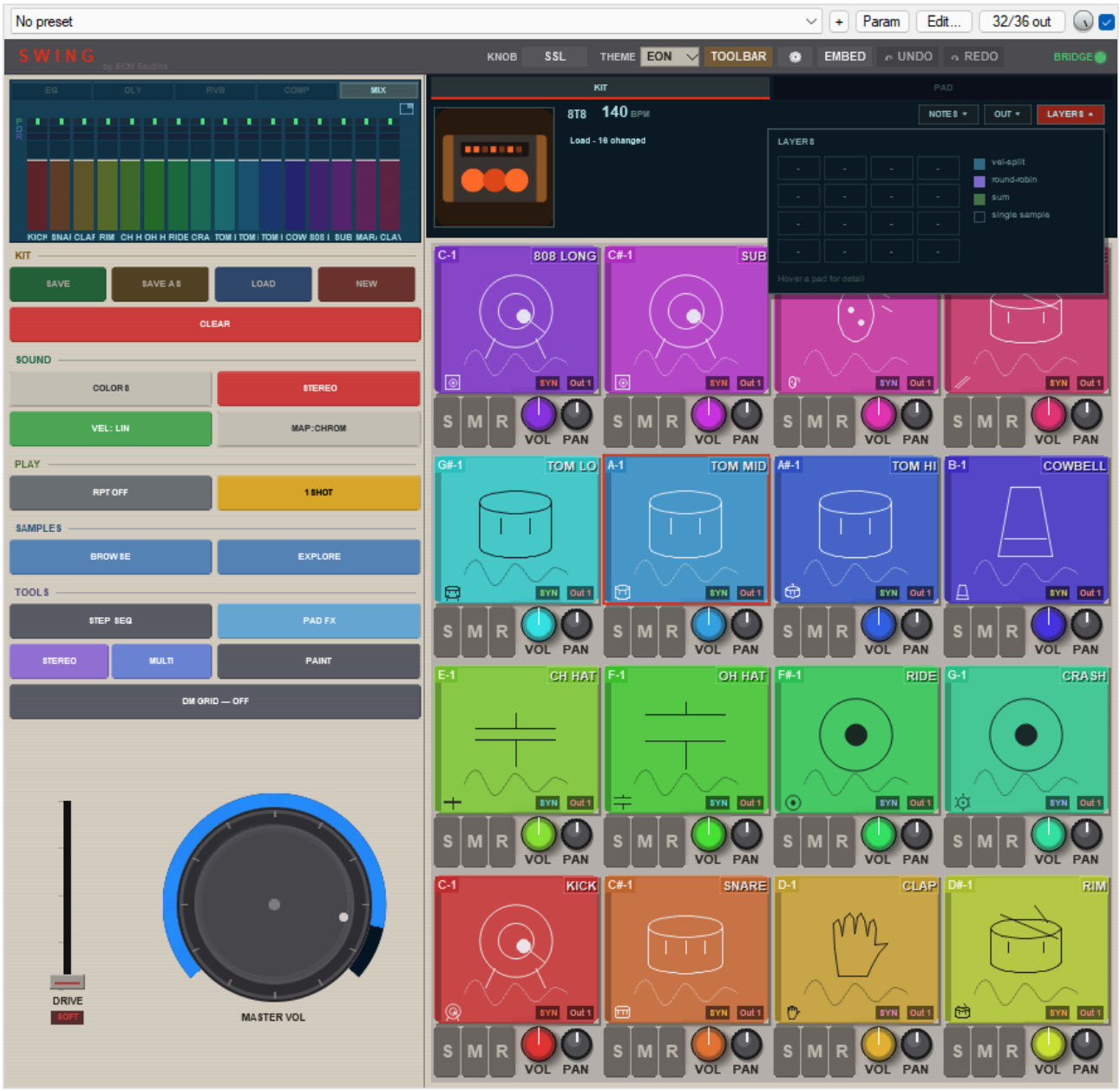
Companions. The console's pad strips, the pad editor and the pickers work on the bank on screen. The Step Sequencer plays any bank by note; its pad badges name the bank on screen. The Drum Matrix lanes and the Drum Strips belong to bank A: the matrix pauses its sync while another bank is on screen and resumes when you come back, and its builds ask for bank A. Kit macros reach all four banks — ALL means

all sixty-four pads, a selected-pad macro follows the selected pad in any bank, and the macro editor's pad row picks pads of the bank on screen.

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6. Layers

Each pad supports up to 4 layers. Layers can be used for velocity dynamics, round-robin variation, or stacked sounds.



Velocity splits. Set each layer's velocity low/high thresholds so different samples trigger at different hit strengths.

Round-robin. Cycle through layers on consecutive hits to reduce the machine-gun effect on fast rolls.

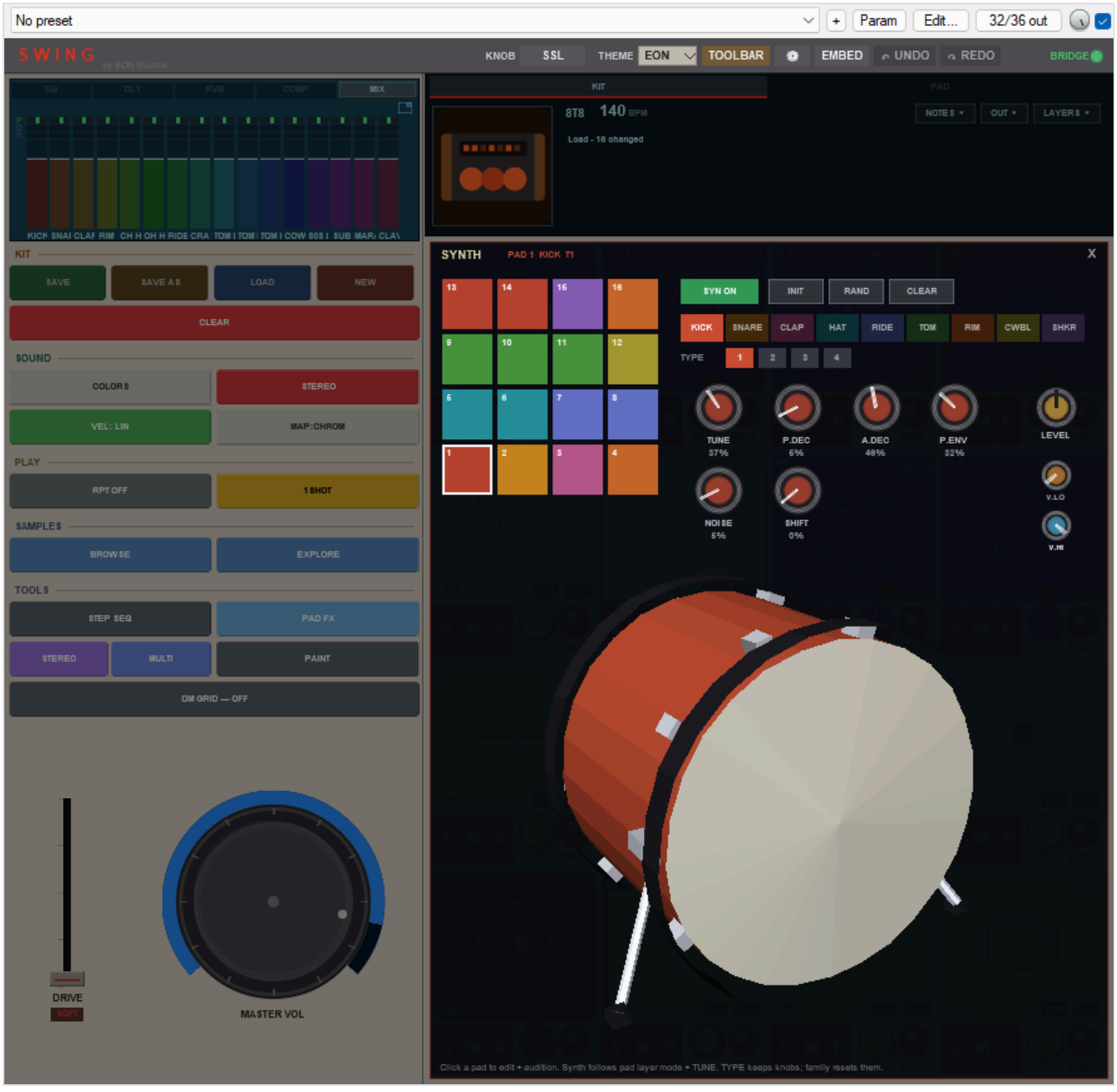
Sum mode. Play multiple layers together. Use tightness to control whether layers align perfectly or smear slightly for thickness — a little smear is how you widen a clap.

Per-layer trim and gain. Each layer has its own start/end trim and gain compensation, so you can level-match sources that were never recorded together.

7. The Drum Synth

New in v3.

Every pad carries a synthesised voice alongside its sample layers, or instead of them. **Nine engine families, thirty voice types**: kick, snare, clap, hat, ride, tom, rim, cowbell and shaker.



Pick a pad on the left, shape it on the right. The macro controls change with the voice — controls that do nothing for a given engine are hidden rather than greyed out, so what you see is what that voice responds to.

In the full window the synth editor is the **SYNTH page of the pad panel** (chapter 8): a pad's SYN chip, Alt+click on a pad or the right-click on a synth-only pad opens the panel on that page, at the side of the pads or below them, and the SOUND and SYNTH chips at the panel's top flip between the pad's sound

editor and its synth. Below the pads the drum sits to the right of the controls. The small faces — the rack face with the editor placed below, and the embeds — keep the full-page view over the pads.

Above the controls sits a 3D rendering of the drum you are building. It is not decoration: it follows the macros and responds to velocity, so tuning the shell changes its shape and hitting harder moves it.

These are ports of hardware-measured references rather than approximations, so the classic voices behave the way the machines they came from behaved.

A synth pad is a full pad. It plays from MIDI and the step sequencer, takes the per-pad EQ, FX, choke groups and sends, saves into kits, and undoes like anything else.

The eleven Vintage Synth kits are built entirely on this engine — 8T8, 9T9, 6T6, CR-7T, SDS, DR-1, POPS, SPACE, TRAP 8T8, DD-101 and PERC JAM. No samples, so they load instantly, weigh almost nothing, and every voice stays editable.

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8. Per-Pad FX

Swing gives every pad its own FX chain. **PAD FX** in the TOOLS group opens the **pad panel** for the pad you clicked last — a pad's right-click opens it too. The panel has two pages, picked by the chips at its top: **SOUND**, the whole editor top to bottom — the pad's colour, its layers and mode, the waveform with its trim, envelope and knobs, the EQ curve with its draggable nodes, the filter, drive, crush and comp knobs with their mode chips, the sends, and the buttons row — and **SYNTH**, the drum synth (chapter 7). By default the panel sits at the side of the pads, beside the browser when the window has room for both (as tabs until it does); the window widens for it as it does for the browser, and takes the rest from the pads where the screen allows only part of the growth. *Pad editor below the pads* in the sequencer's OPT menu moves it under the pads, where the sound editor spreads into two columns and the synth's drum sits beside its controls; the strip's chevron, drag handle and a PAD FX chip serve it there as they serve the sequencer, and the panel opened last is the one shown when two are sent below. Each side panel has its own X; the chevron at the window's edge folds them all. The seam between the pads and the side panel is its width handle: drag it left for a wider panel, right for narrower, wheel for small steps, right-click for the panels' own widths; the pads give the room and the window stays as it is, and the width is saved with the project. When the window is too short for the editor beside the pads, the panel below gives up height first. Right-click **PAD FX** for the floating Pad FX window. The small faces keep the editor over the pads.



HPF / LPF. High-pass and low-pass filters (20 Hz–20 kHz) for cleanup and tone shaping.

EQ. With an interactive curve — drag the nodes. Start with subtle ±1–3 dB moves.

Saturation. Five drive modes: Soft, Tube, Tape, Fuzz, Clip. Use small amounts first; Clip is the most aggressive.

Bitcrush. Rate reduction plus bit depth, for lo-fi textures.

Delay / reverb sends. Send each pad into the global engines for cohesion.

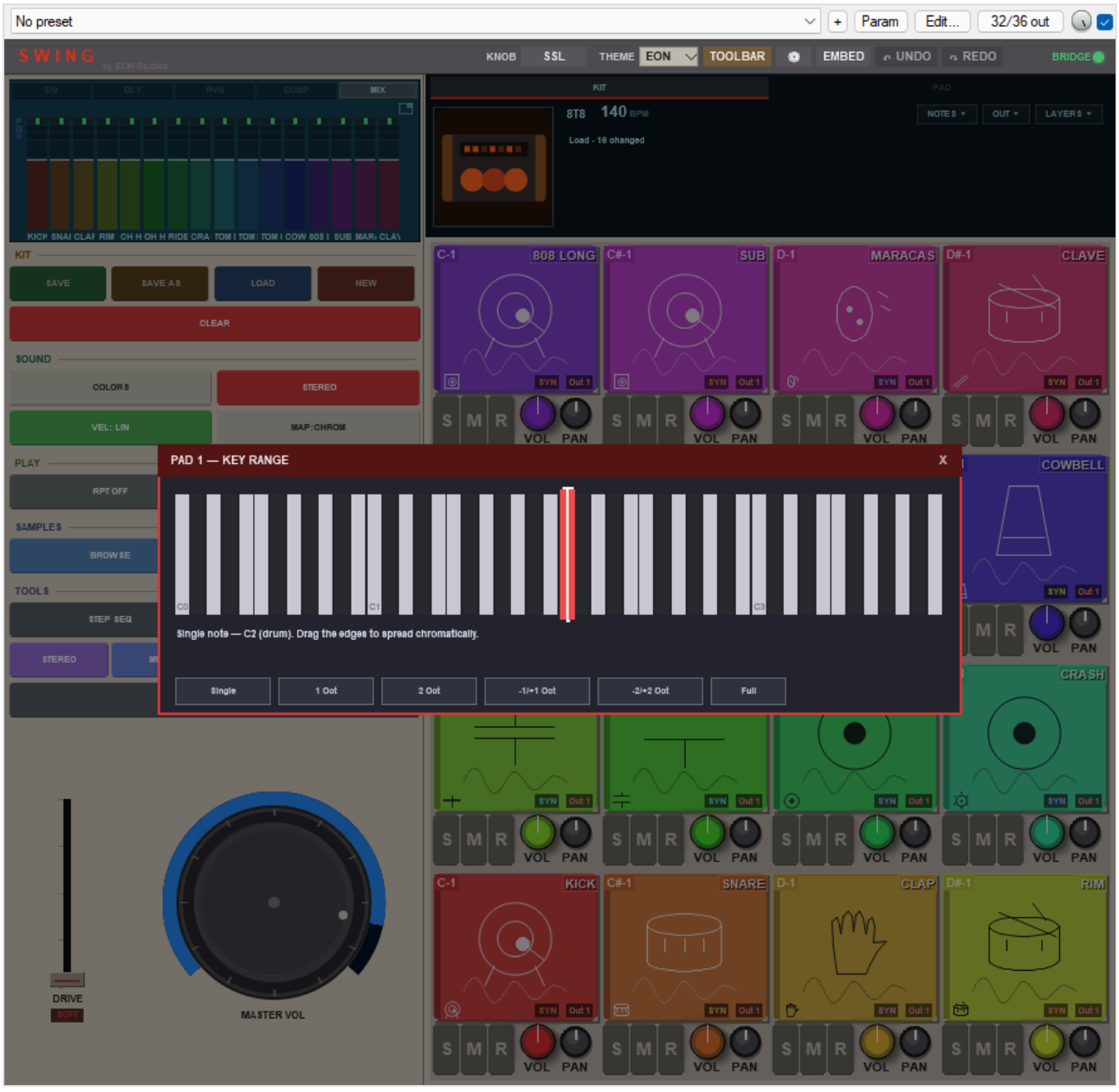
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9. Key Ranges & Tune Modes

New in v3.

Key ranges

Give a pad a key range and it plays across the keyboard instead of on one note.



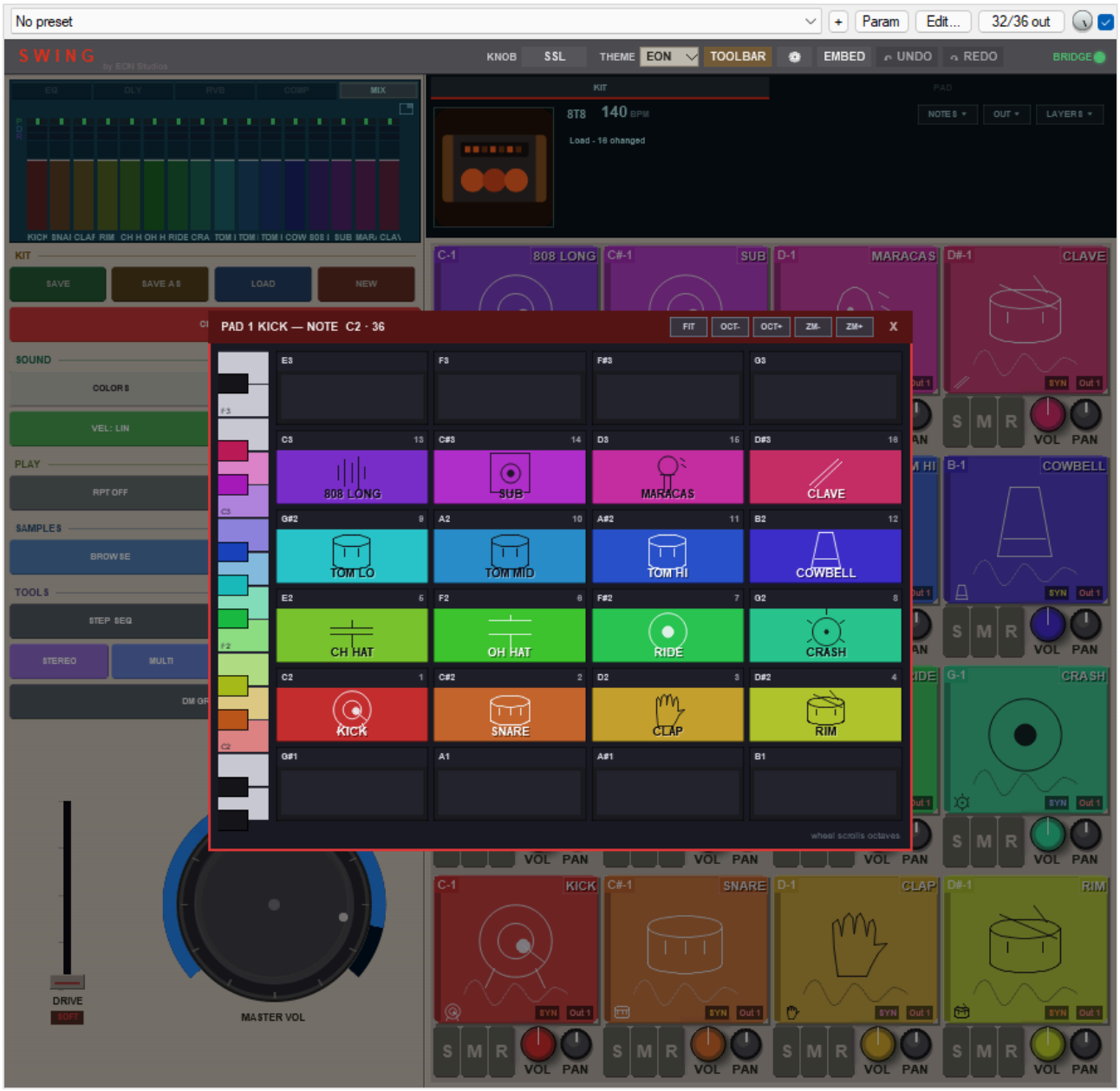
Right-click the **RANGE** pill to set how far the range reaches either side of the pad's own note. The range is stored as offsets from that note rather than as fixed keys, so it survives a kit change — load a new kit and your playable pad is still playable.

Tempo-repeat follows the key you are actually holding, so held-note rolls track the melody rather than hammering the pad's root.

Two limits worth knowing. A ranged pad is **monophonic** — it plays melodically across the keys, one note at a time, and does not play chords. And key ranges track in **Repitch** mode only; Tuned and Stretch support is still to come, and Swing says so in the range editor rather than leaving you to find out.

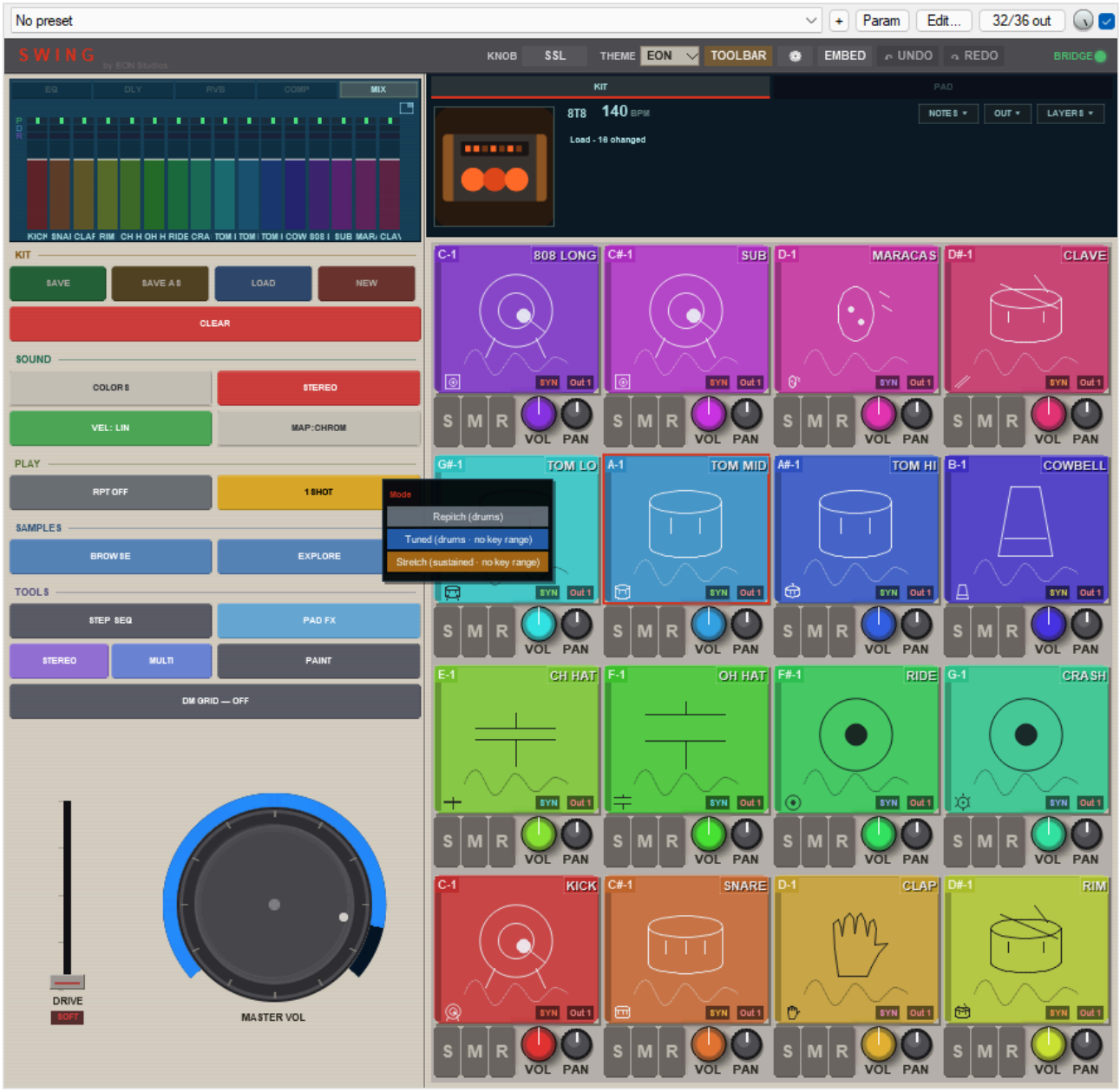
The note picker

Click the **NOTE** pill on a pad to open the note map framed on that pad.



Click a key to set the pad's note, or drag the pad's chip along the keyboard. It opens framed on your kit with an empty row above and below, so you can see where the pad sits before you move it.

Tune modes



Repitch is varispeed — pitch and length move together, like tape. Free, and right for drums.

Tuned is pitch-preserved, using a bake with per-layer anchors so velocity-layered pads pitch every layer correctly.

Stretch is real-time time-stretch, and requires the EON engine.

The hints in the picker steer sustained material toward Stretch and drums away from it. **Without the EON engine installed**, Tuned and Stretch play as Repitch, and Swing shows a notice rather than quietly sounding wrong.

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10. The Big View

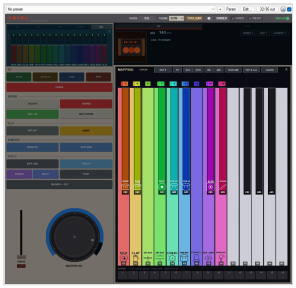
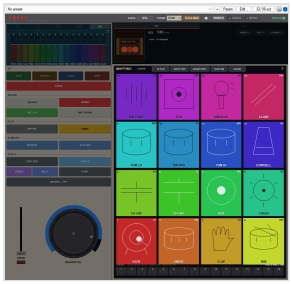
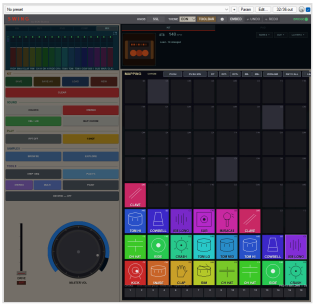
New in v3.

Ten views open over the pad grid, all bindable to keys and toolbar buttons, plus an eleventh action that closes whichever is open.

Mapping

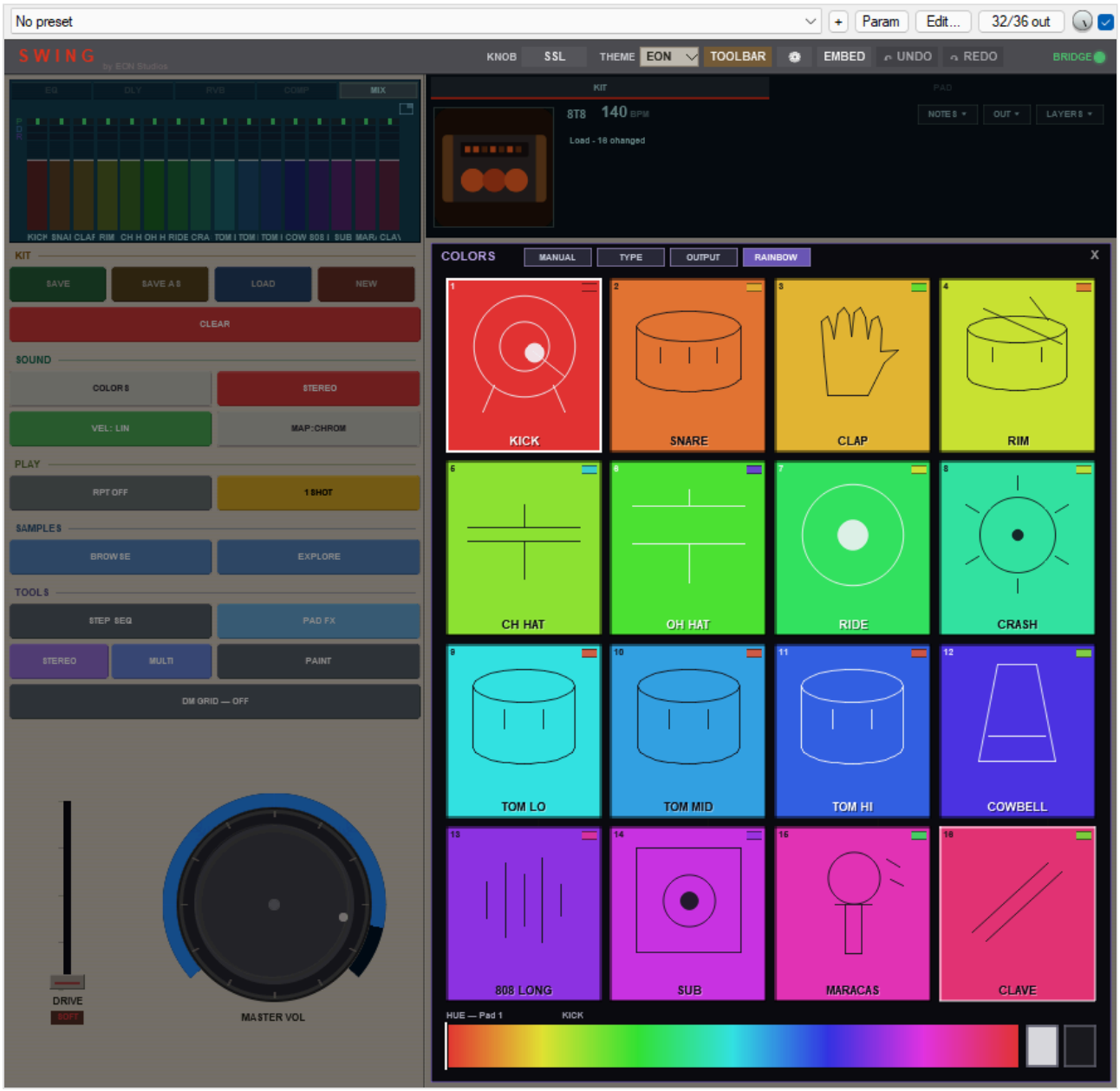


A piano-roll note map with draggable keys. Seven layouts cover the usual controller shapes:

Keys	8-pad	16-pad	Ableton Push
			

Bank A/B switches the lower and upper halves. The **choke lane** underneath is editable — that is where open and closed hats get grouped.

Colours



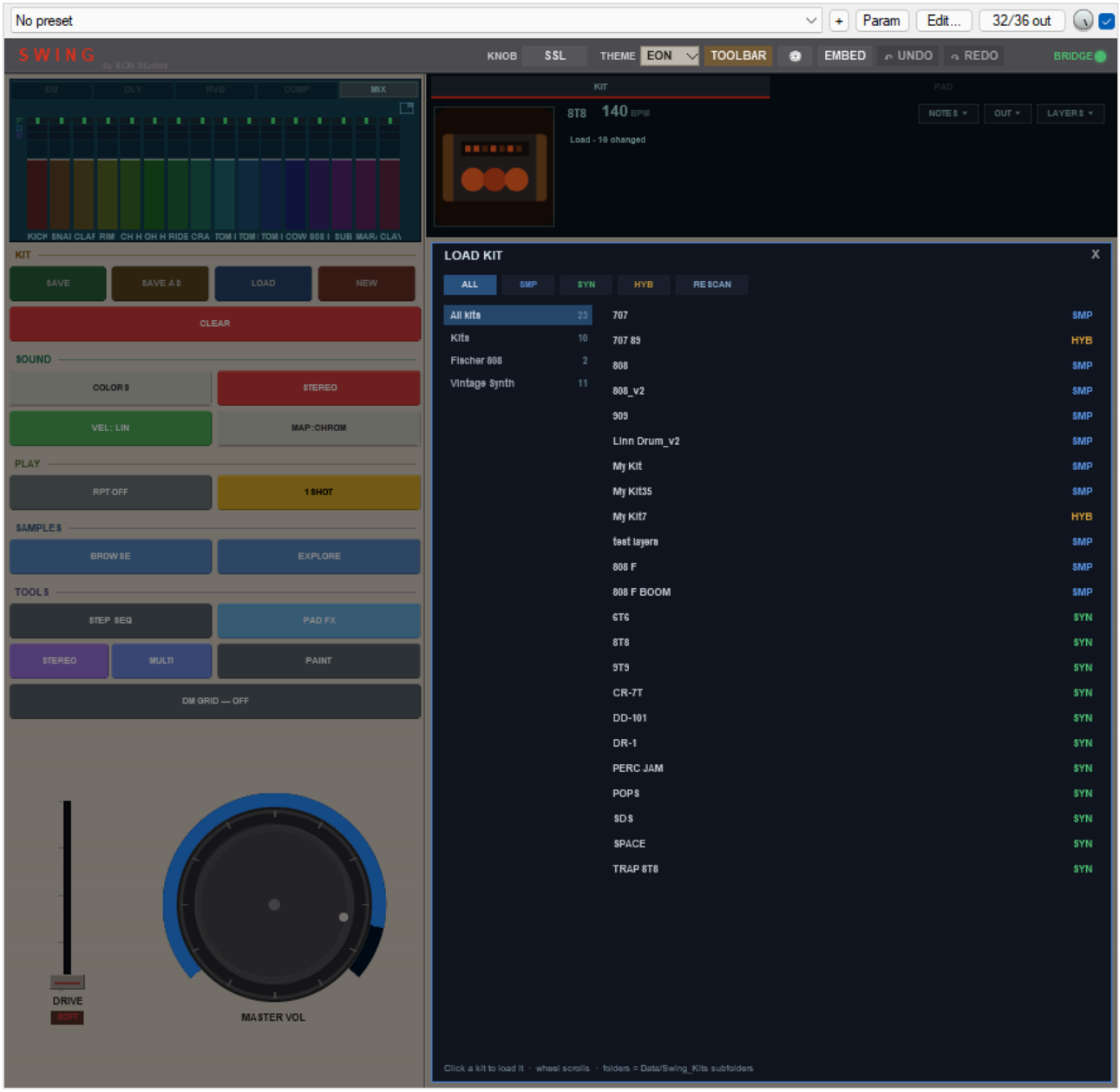
Four modes — MANUAL, TYPE, OUTPUT and RAINBOW — plus a hue strip for the selected pad. Pad colour carries through to track colours and sequencer lanes, so a kit stays recognisable everywhere it appears.

Outputs



The routing matrix for the bank on screen: bank A's sixteen Outs, or a bank's own track and the pads split off from it, with *n of 10 in use* for the own outputs. See chapter 15 for building a multi-out rig.

Kits



The library, browsable by folder and filterable by sample, synth or hybrid. Left-clicking **LOAD** comes here.

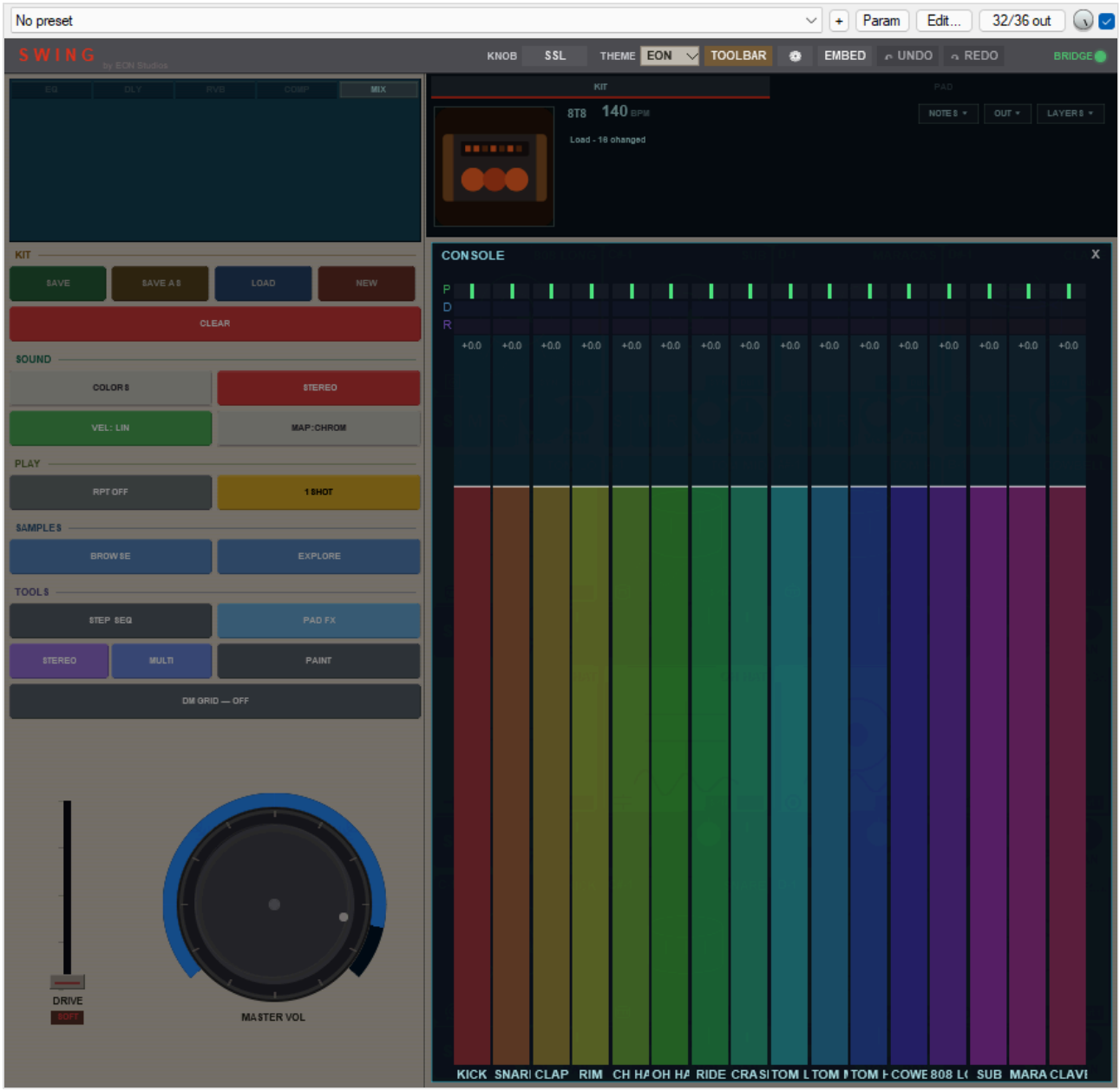
> **Coming from 2.1?** The old *Load Kit (.swing)* entry is gone from the LOAD

> right-click menu. Loading a kit file lives in the Kits view now.

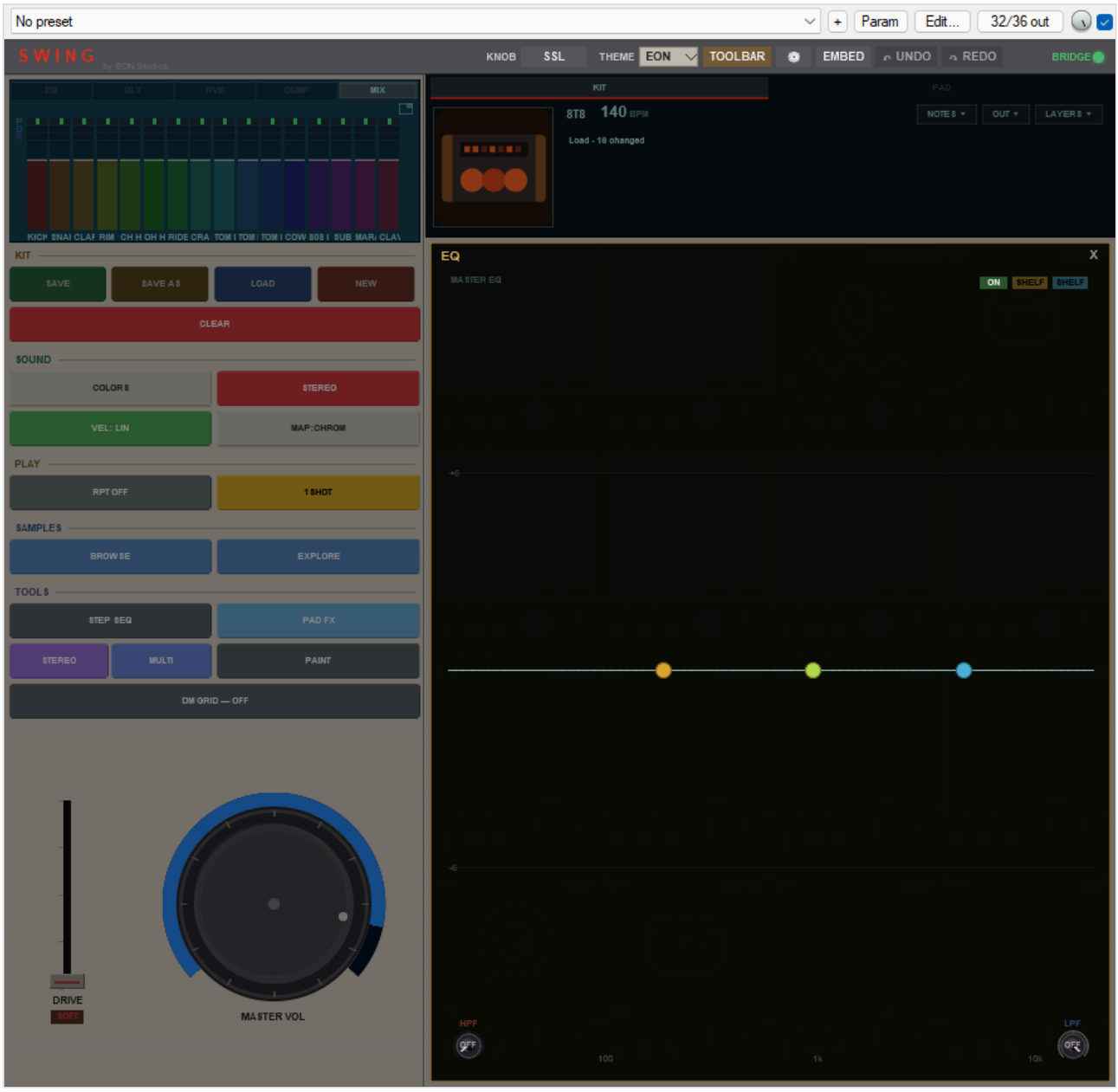
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11. Global FX & Mixer

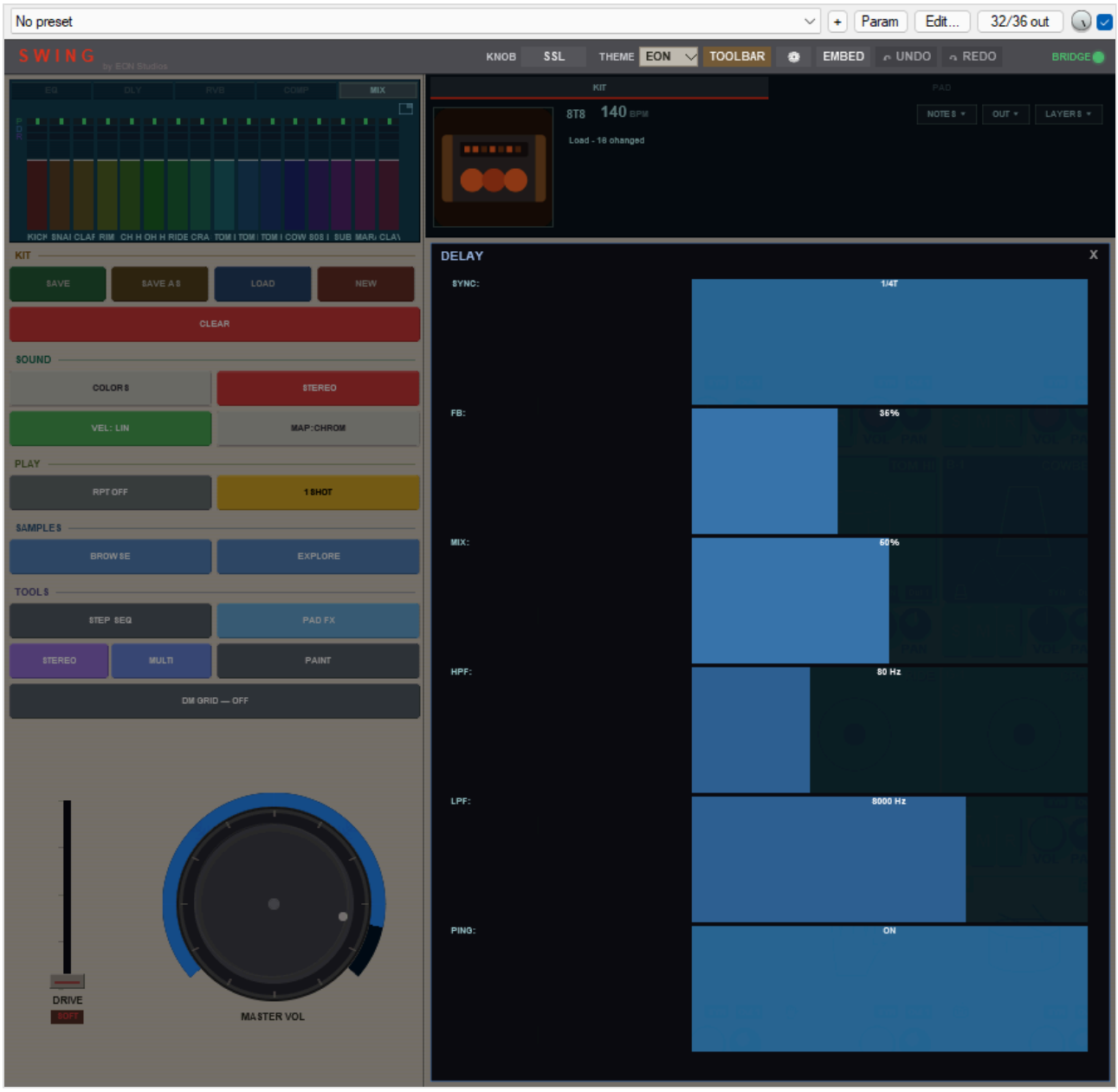
Use the LCD tabs for master processing.



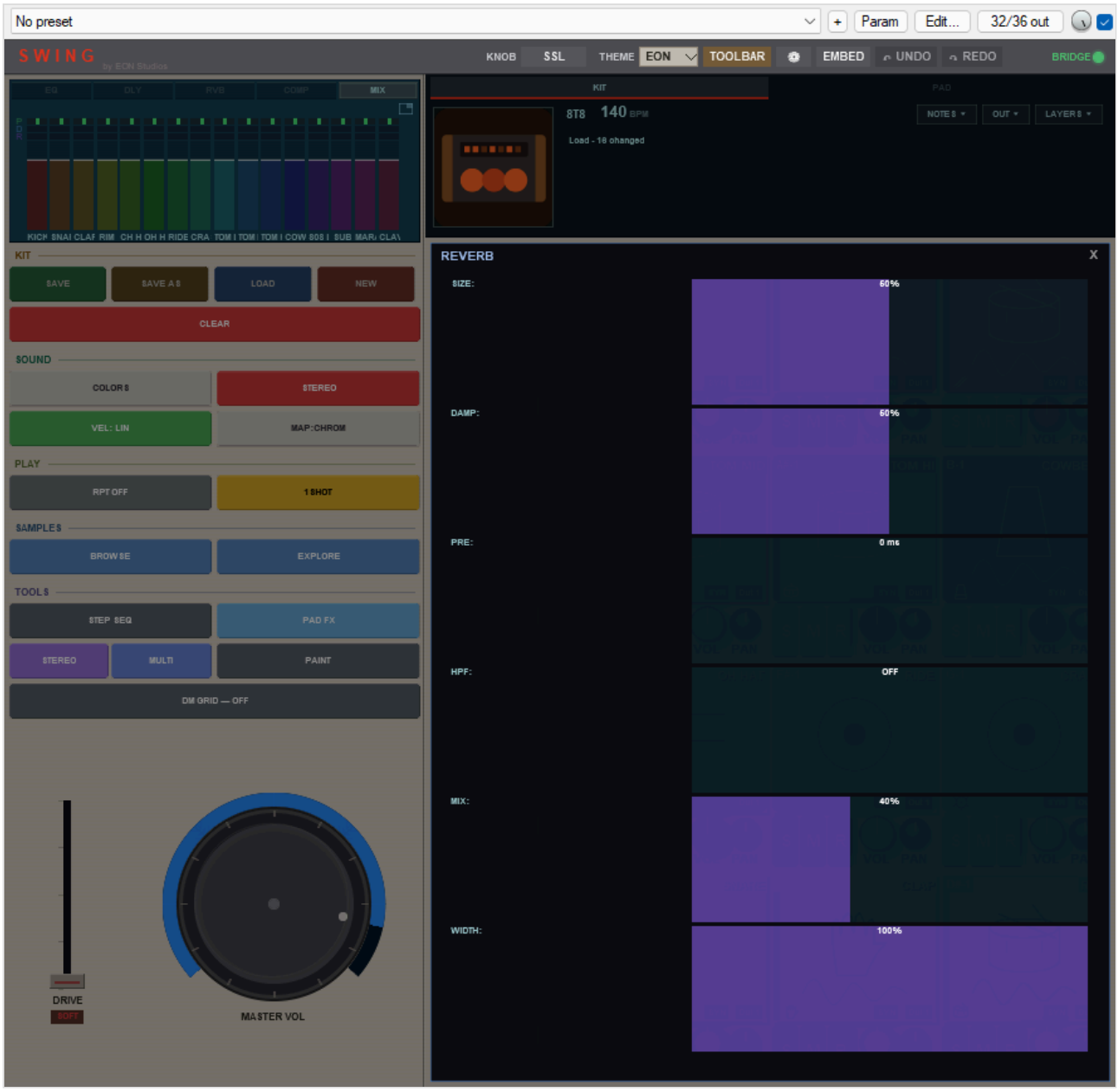
EQ. Four bands of zero-zipper state-variable filtering, with a live curve. Rebuilt in v3 — coefficients glide instead of stepping, so sweeping is silent.



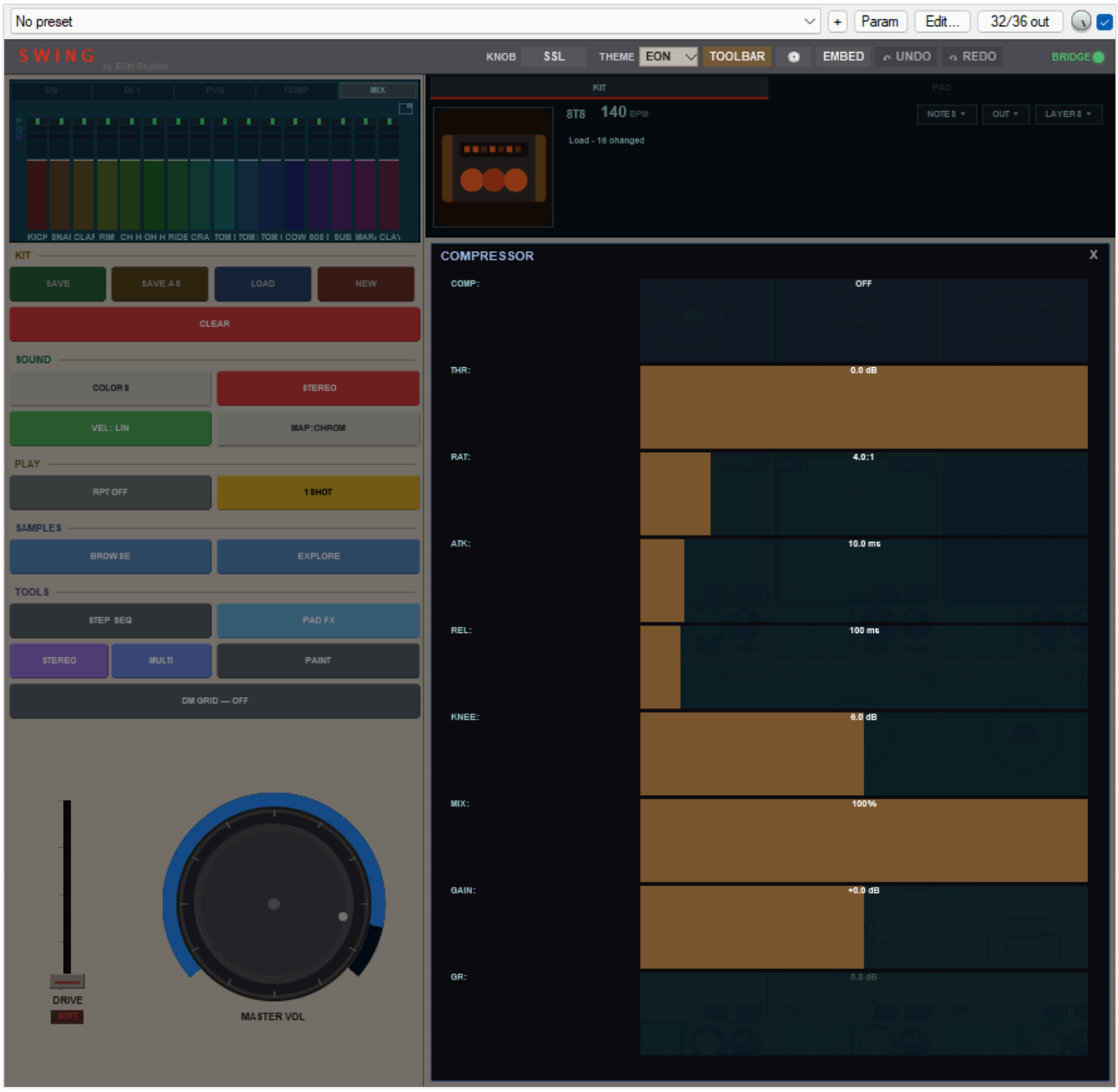
Delay. Tempo-synced, with feedback and tone controls.



Reverb. Size, damping, width and pre-delay.



Compressor. Threshold, ratio, attack, release, knee — glue and punch.



Limiter / master volume. Keep peaks under control, and level-match when comparing presets.

12. Swing Kit Bridge

The Kit Bridge is a background Lua service that enables kit save/load and other operations JSFX alone cannot do safely. The Windows installer registers it to start with REAPER. On a ReaPack install, run **Swing_Kit_Bridge** once from the Action List — it registers itself and auto-starts from then on.

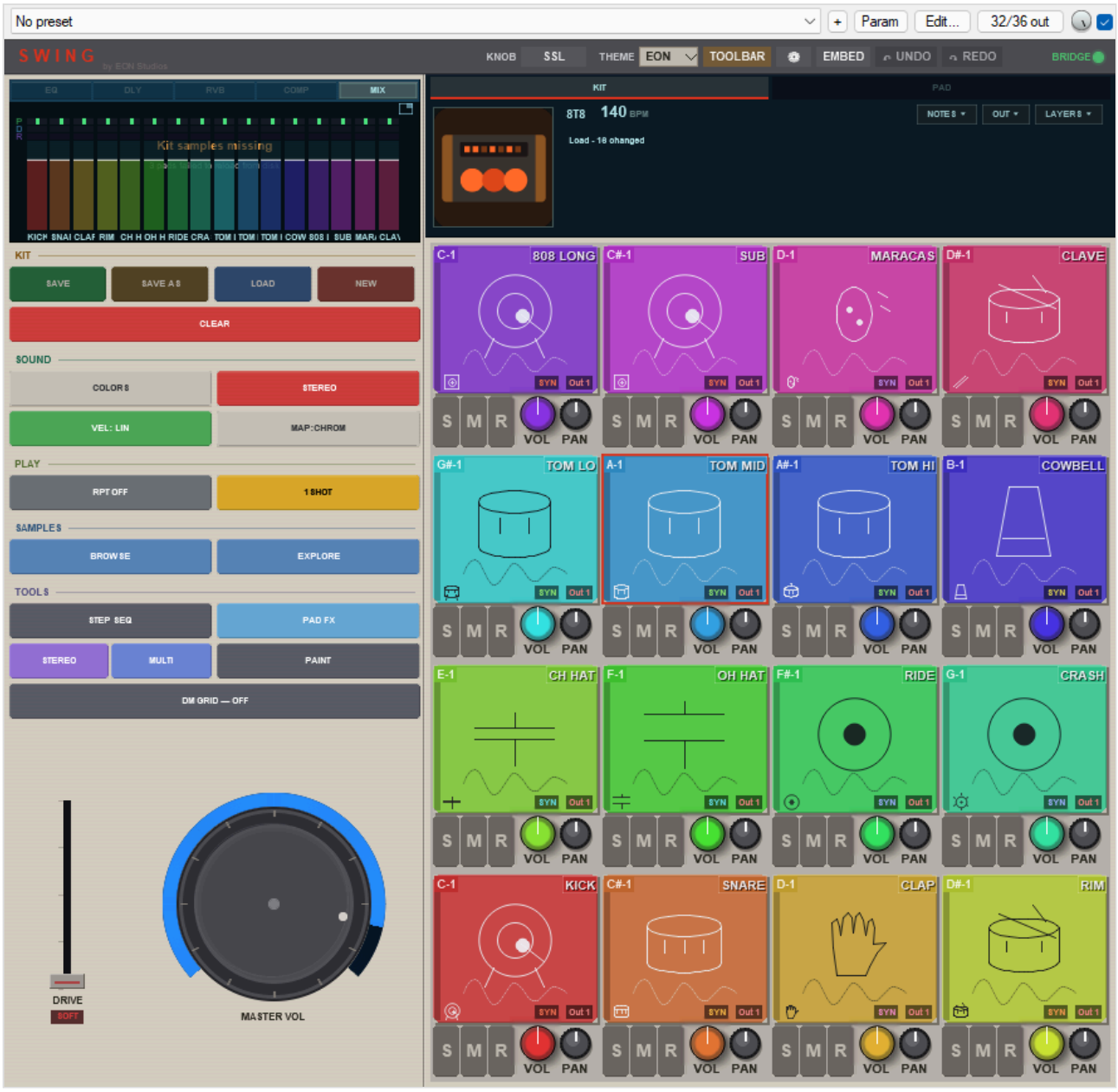
Save/Load kits. Writes and reads `.swing` files, including pad names, colours, output routing, ADSR, FX settings, synth voices and cover artwork.

Batch import. Load a whole folder of samples and auto-distribute to pads.

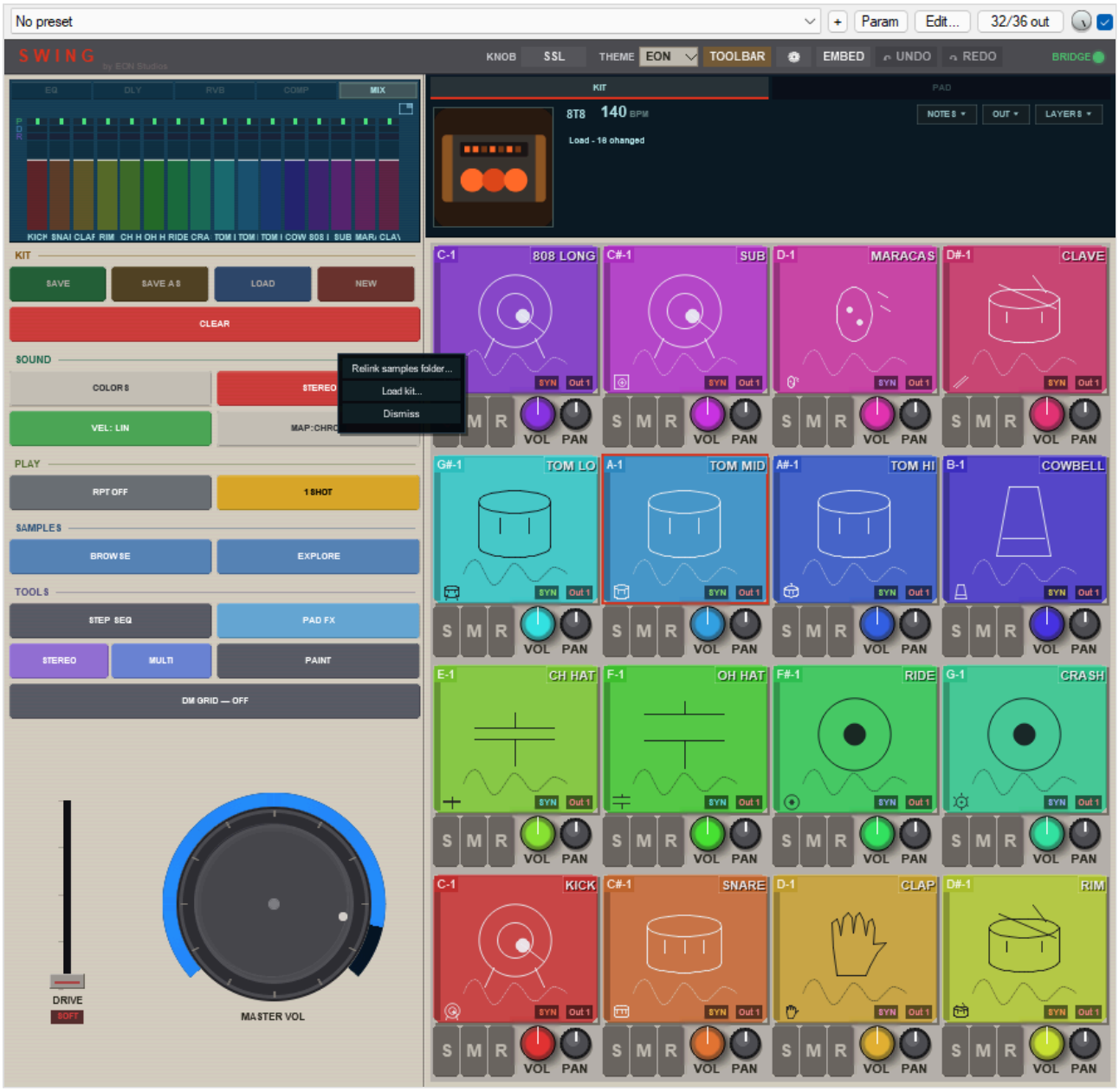
Chop to pads. Slice an arrangement selection straight onto pads, from the action list or the LOAD right-click menu. The chopper can also export its slices as WAV files.

Export. Beyond `.swing`, a kit can leave Swing in two directions: **Export SFZ** writes a self-contained SFZ — the WAVs travel with it — keeping tune, volume, pan and note. **Export RS5k** converts the kit to a track of ReaSamplOmatic5000 instances (velocity layers, round-robin and containers included), and the converter is two-way: **Import RS5k** pulls an existing RS5k rig onto pads.

Relinking. *New in v3.* When samples go missing — you moved or renamed a folder — Swing says so instead of leaving silent pads.



Clicking the banner offers to relink the folder, load a different kit, or dismiss.



If a whole project moves, Swing re-points automatically.

Embed audio: Auto / Always / Never. *New in v3.* Decides whether pad audio is written inside the project file. *Auto* keeps projects small; *Always* makes a project self-contained and portable, at the cost of size.

Housekeeping. Pad audio that was never saved into a kit or project, and has gone untouched for 30 days, is deleted automatically so the scratch store does not grow without limit. This only ever touches Swing's own unsaved scratch audio — kits, projects and your own sample files are never involved.

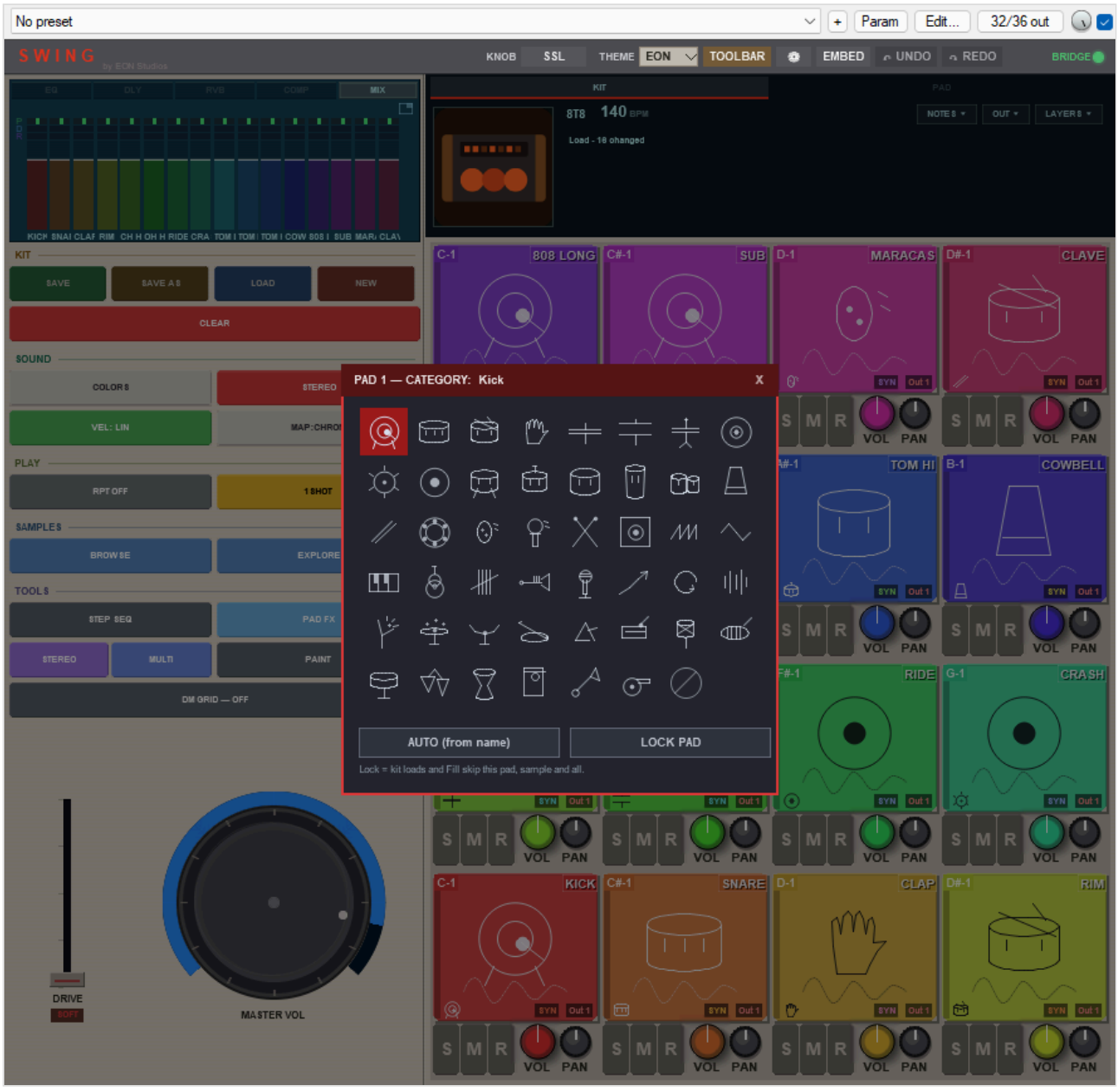
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13. Kit Categories & Artwork

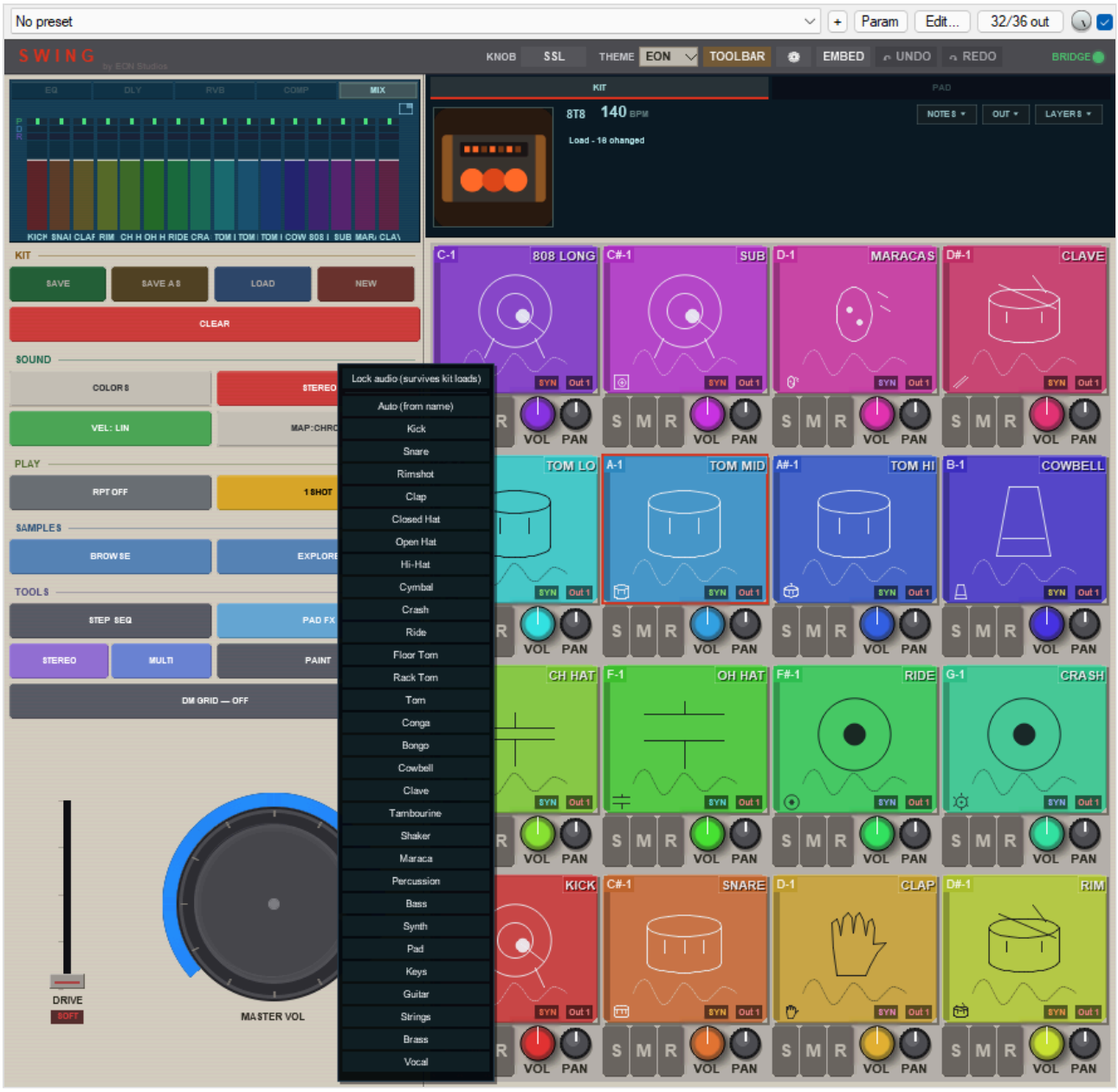
New in v3.

Categories, LOCK and ADAPT

Each pad carries a category glyph in place of its number, picked from a grid of 46.



The same vocabulary is available as a list, with LOCK at the top:



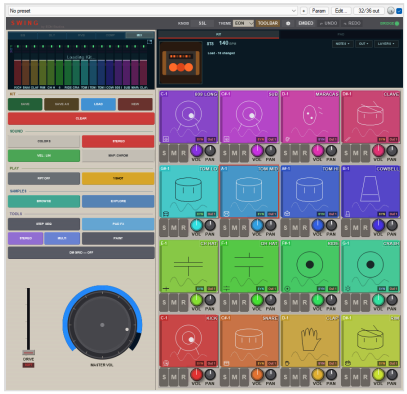
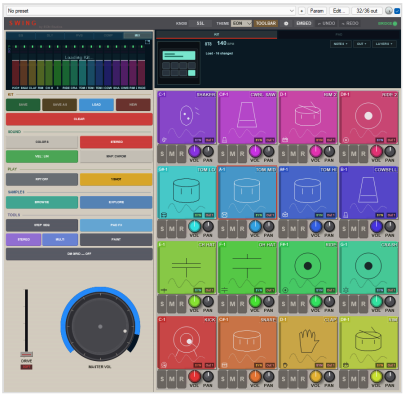
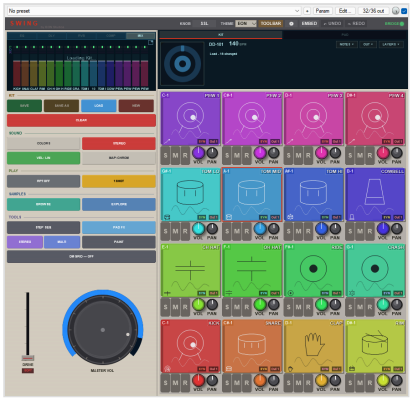
LOCK keeps a pad exactly as it is through kit loads — locked pads are untouched by anything a new kit brings.

ADAPT re-aims your existing patterns at the new kit rather than leaving them pointing at the old one. A change map shows what moved after a load.

Kit artwork

Kits carry a cover image inside the `.swing` file, so it travels with the kit wherever the kit goes. Drag a **PNG or JPEG** onto the KIT tab's cover tile to attach one. The new cover appears immediately, but it is only written into the `.swing` file when you next save the kit.

All thirteen factory kits ship with machine-portrait covers.

8T8	DD-101	SPACE
		

14. Swing Browser

The Swing Browser is an ImGui-based sample browser with folder navigation and audio preview. The mini browser inside Swing's window sits at the side of the pads, or below them when the sequencer's OPT menu says *Browser below the pads* (chapter 16).



The sidebar. Five panels, each collapsible:

- **Pads** — a mini pad grid with the selected pad's name and note (editable from the dropdown), and a **LOAD → PAD** button for click-loading without a drag.
- **Kits** — the kit list, with **Fill my layout** to pour a kit into your current pad arrangement.
- **Categories** — the browser auto-categorises what it finds (kick, snare, hats, toms, percussion and more) and shows live counts per category. Click one to filter the file list.

- **Shortcuts** — Desktop, Downloads, the project directory, REAPER resources and drive roots. **Favorites** — right-click any folder to pin it. **Recent** — where you've been.

The file list. Columns for name, detected category, type, channels, BPM, key, size, duration and detected root note — click a header to sort. The path bar navigates like a file manager, **Subfolders** searches recursively, and Ctrl+click / Shift+click select multiple files. Search filters as you type.

Preview. Audition before loading: waveform display with **Spectral** and **Grid** overlays, transport, volume, pitch offset in semitones, **Loop** and **Auto**-play on selection. Preview playback needs SWS installed.

Drag & drop. Drag files directly onto pads — including onto a Swing embedded in the track panel or mixer. Multi-file drags fill consecutive pads.

Pad actions. Right-click a pad in the sidebar for mute, solo and reverse toggles, **Set Trigger Note**, and **Open Targeted FX in REAPER**.

Targeting. In projects with multiple Swings, the browser targets a specific instance — the title bar and the **[Track N]** chip say which — so a dropped sample lands where you meant it to.

Looks. The **EON** button picks the browser's own theme — fourteen of them, from the EON house look to REAPER-native, console and DAW styles.

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15. Multi-Out Routing

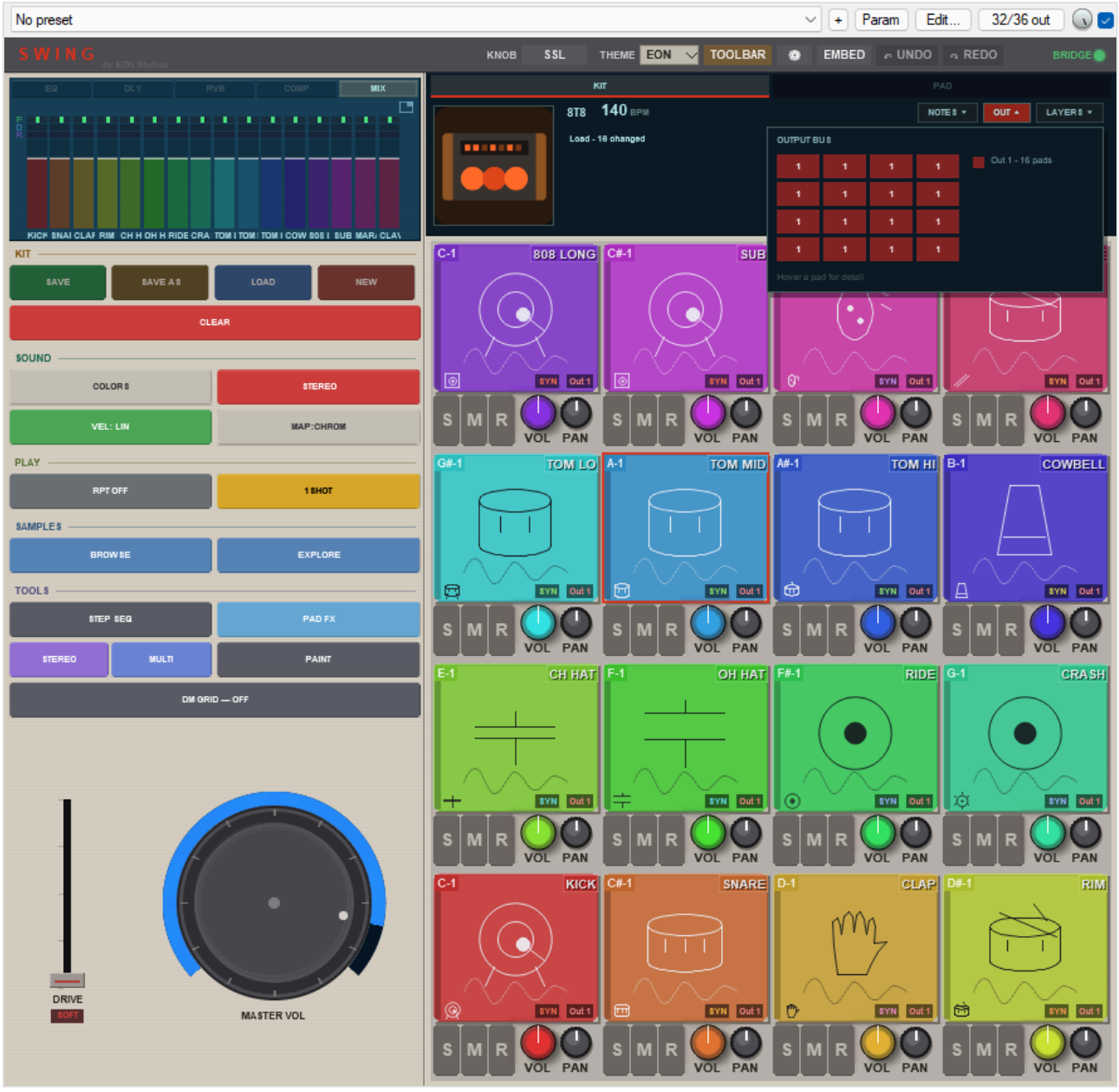
Swing's pads leave three ways. **STEREO** sums everything on channels 1-2. **MULTI** gives each pad, each bank and each own output its own REAPER track. **WIRED** keeps everything on Swing's own track and lets you put any plugin on any pad. The mode button in the left panel — it reads **STEREO**, **MULTI** or **WIRED** — toggles STEREO / MULTI; right-click it for the builder, Wired and the conversions.



Build the rig. Right-click the **STEREO / MULTI** buttons in the tools row and choose **Build Multi-Out (Audio)**, or pick **Build Multi-Out Tracks** from the mode button's menu. That creates the matching REAPER tracks so each pad appears on its own mixer channel. The build reads bank A: Swing's own button shows bank A first, and a build started from an action while another bank is on screen refuses with a notice until you show A.

The same menu offers **Build Stereo Grid**, **Build Drum Matrix (MIDI lanes)** and **Build Both (Audio + MIDI)**.

The KIT tab's **OUT** panel shows all sixteen buses at a glance:



Built, it looks like this — a channel per pad, each carrying its own Drum Strip, with the pad's colour and category icon carried through to the track:



What the build makes. A folder named after the kit holds the Swing track and an **Audio** folder. Inside Audio: a **Bank A** folder with the sixteen pad tracks, a track per bank B-D that has sounds, the own-output tracks inside their bank's track, and the FX returns at the end. The Audio folder itself carries Swing's master section as three plugins — **EON Tone** (the five-band EQ and drive), the bus compressor you picked at build, and **EON Output** (master volume and the limiter), which follows Swing's master fader and limiter so there is one place to set them.

Banks B-D. A bank B-D pad plays through its bank's track — *Bank B: 707* — and effects on that track process the whole bank. Pick **Own output** from the pad's Out menu (on the pad tile, in the pad editor or with the LCD's Out stepper) and the pad gets a track of its own inside the bank's folder, named after it — *C3: Snare*. Up to ten pads across banks B-D can have their own; past ten the entry greys out with *all 10 in use*. The track appears by itself within a second, as long as the pad has a sound on it. Bank A's pads keep Out 1-16 and can also be sent to a bank's track. A track you delete by hand is not remade behind your back: run the build again and choose Update.

Update and Rebuild. Building again over an existing rig asks which you want. **Update** keeps the tracks, refreshes names and colours, and adds whatever is missing — a new bank's track, an own output's track, the Bank A folder on a project built before banks. **Rebuild** deletes Swing's own tracks and makes fresh ones; it refuses, naming the track, while an old one still carries your own plugins, recorded items, automation or routing, so nothing of yours is lost without you seeing it. Both leave your own tracks and your own sends alone.

Swing's panel in the mixer. Once the Swing track has more than 38 channels — the first bank track or own output does that — REAPER 7's default theme narrows its mixer panel to half width. Swing asks once: **Use Layout B** keeps the panel full width by putting Swing's mixer strip on the theme's Layout B (which also changes any other tracks you have put on Layout B); **Leave it** leaves the theme alone. Change your mind later with the gear menu's **Mixer panel full width (Layout B)** row. Other themes are never touched; one that narrows the panel gets a single line in REAPER's console, once.

Mixing. Once built, each pad, bank and own output has its own channel in REAPER's mixer — add EQ, compression and sends per drum there while keeping Swing's pad engine simple.

FX returns. *New in v3.* Opt-in delay and reverb return outputs for multi-out rigs, with per-pad send knobs on the Drum Strip — and a third, **Smash**: a per-pad SMASH send next to DLY and RVB, feeding a parallel-compression return driven by **EON Weld**, a bus compressor with a real gain-reduction needle. The build can also drop a Weld on the drum bus itself for glue, with gentler defaults than the Smash instance.

Wired

Since 3.0.3.

Wired puts any plugin on any pad, on one track. Each pad leaves Swing on its own channel pair, runs through whatever you put on that pair, and an **EON Patchbay** plugin at the end of the chain brings it back into Swing ahead of the master EQ, compressor and limiter. One track, any third-party plugin on any pad, one buffer of latency that Swing reports as PDC. On REAPER 7 each pad gets its own named FX container, so a plugin dropped in through REAPER's own browser is routed the moment it lands. In the console, click an insert slot and put a plugin on that pad; the picker does the rest.

Switching. The **WIRED** button in the tools row switches it on and off. Right-click the mode button for the **tap point** (on the channel, or on the bus) and the two conversions. **Wired -> Multi-Out** moves each pad's plugins onto its own track, in order, a bank's onto the bank's track, an own output's onto its track, and the master inserts either side of the Patchbay onto the Bank A folder — except the ones the other conversion brought from the Audio folder, which go back there. **Multi-Out -> Wired** brings them all back, fader included; the Bank A folder's plugins and the Audio folder's own go after the Patchbay as master inserts, where they process the whole output. It refuses, naming the track, while a track it would remove carries something of yours.

WIRED? in amber means the Patchbay is not answering: it must be last in the track's FX chain and enabled. Until it is, pads sum inside Swing as in STEREO and your pad inserts are silent.

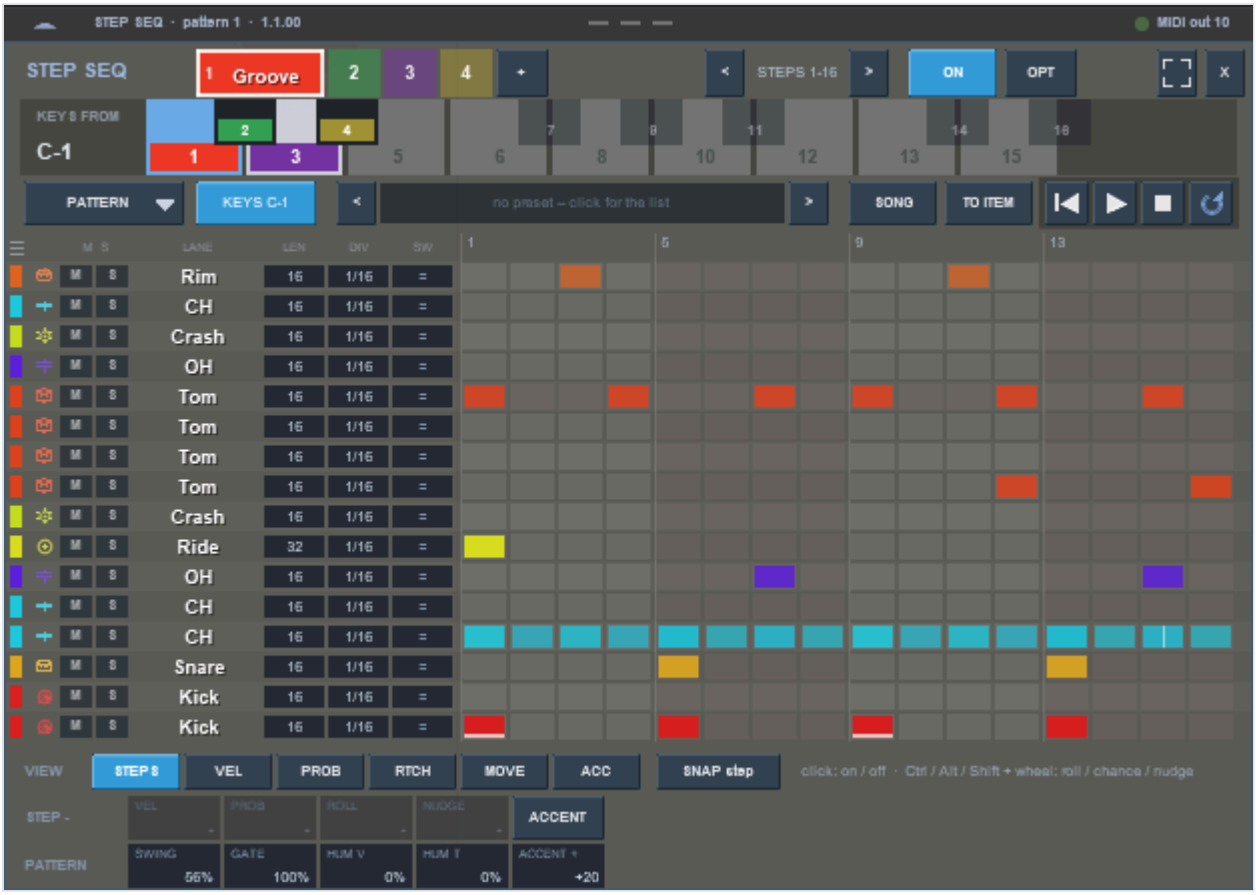
Banks under Wired. Bank A's pads leave on the channel pair of their output (Out 1 = 3-4 … Out 16 = 33-34), banks B-D on their bank's pair (channels 39-44) and own outputs on 45-64. Every pad keeps its own pan and its own delay and reverb sends. A bank's pads share one insert chain, the *Bank B* container; an own output has its own, *C3: Snare*. A pad on a bank's pair leaves after its pan whatever the tap point. The Patchbay's embedded meter shows all 29 lanes — bank A's 4x4, Bank B / C / D, Own 1-10 — all the time. If the Patchbay in the chain is older than this Swing, every pad leaves on bank A's sixteen pairs as before, and the OUT button's tooltip says so.

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16. The Step Sequencer

New since 3.0.19.

Swing has a step sequencer of its own, in a panel under the pads: sixteen lanes, one per pad, thirty-two patterns, and REAPER's timeline as the clock. The grid is the project. Step 1 of a lane is beat 1 of the project, so a pattern sits in the same place however playback started, and a lane of seven steps runs against a lane of sixteen the way it does on a hardware box. Steppa, the step sequencer that came with 3.0, is a free plugin of its own now (chapter 17); this one lives inside Swing and knows the kit.



Opening it. The **SEQ** button in the TOOLS row (the left half of SEQ | STEPPA) opens the panel; the window grows taller to make room, and the button stays lit while it is open. The arrows button at the right of the panel's toolbar spreads the grid over the whole window and brings the pads back; the **X** closes the panel. The panel takes its colour from the theme. In a docked or embedded face the panel draws and takes clicks but gets no keys.

Lanes. Each lane plays one pad, in any bank. Click the lane's name to aim it at another pad: the picker lists the bank's sixteen pads by name and then the other banks. Click the lane's icon to hear its pad (with the icons hidden, the colour chip does it). The lanes sit bottom-up like the pads, kick at the bottom (OPT turns them the other way). A lane's header carries the pad's category icon in the pad's colour (an option), **M** and **S**, the name, **LEN** (1 to 64 steps), **DIV** (1/4 to 1/32, triplets and dotted) and **SW** (the lane's own swing, or **=** to follow the pattern's). Click a column's label in the ruler to hide the column; the menu at the ruler's left

end brings columns back. Mute and solo hold in every pattern, and while any lane is soloed only the soloed ones play.

Steps. Click a cell to switch a step on or off; drag across cells to paint. The views at the footer's left — **STEPS**, **VEL**, **PROB**, **RTCH**, **MOVE**, **ACC**, or the keys `s v p r n a` — show the same cells with a different value on them: the velocity, the chance (how often the step plays), the roll (two, three or four hits in the step), the nudge (early or late by up to half a step) and the accent marks. The step editor beside them sets the selected cell: **VEL**, **PROB**, **ROLL**, the roll's **HIT 2..4** velocities, **NUDGE**, **ACCENT**. Hold Ctrl, Alt or Shift and turn the wheel over a cell in any view to set its roll, chance or nudge; an off step switches on. A right-click on a cell offers the lot — hear the pad, the roll, probability 25 to 100, nudge early, late or none, accent, clear the step — and *every 2nd, 4th or 8th step from here*. A right-click on a lane's name fills every step, every 2nd, 4th or 8th, or clears the lane.

The pattern's feel sits in the footer: **SWING** (odd steps late, by this much of half a step; a lane's own SW overrides it), **GATE** (how long a gate-mode pad holds, as a share of its step), **HUM V** and **HUM T** (humanize the velocity and the timing; the dice are the pattern's own, so a render plays the same every time) and **ACCENT +** (how much louder an accented step plays, on top of its own velocity).

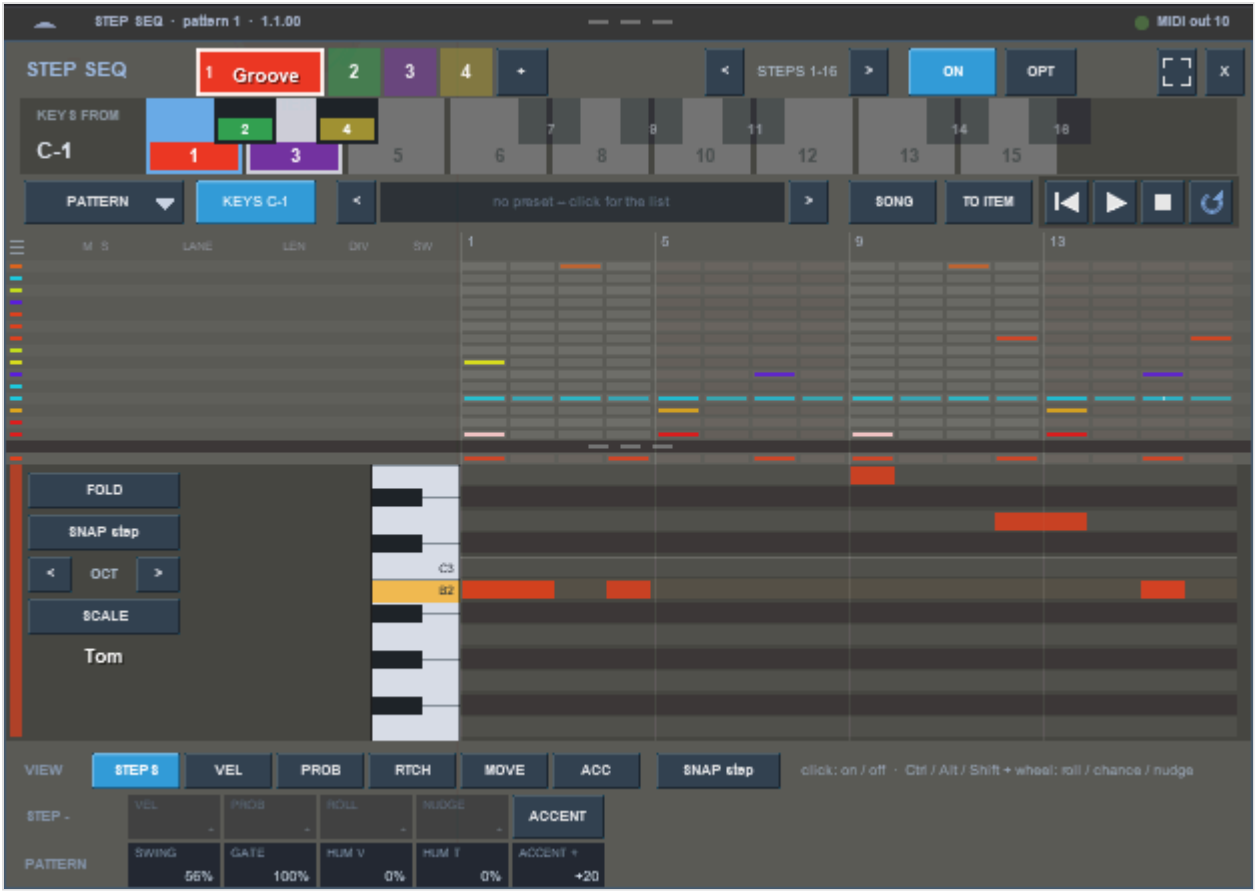
Patterns. Up to thirty-two, shown as coloured chips along the top: a fresh sequencer shows one and a **+** for the next. A chip carries its number and, once the pattern has a name, the name; the one on screen is ringed and a dot marks one with hits. Click a chip to see its pattern; the keys `1` - `8` pick from the first chip shown; when the chips outgrow the row it scrolls (the wheel over it, or the arrows at its ends). The **PATTERN** menu — on the button, or on a right-click of any chip — holds the pattern tools: copy and paste (Ctrl+C, Ctrl+V), duplicate into the next empty slot, halve and double (every lane one division coarser or finer, the hits kept in time; refused whole when a hit would be lost), clear this pattern or all thirty-two, aim the pattern at another bank, rename it (type on the chip, Enter keeps the name), give it a colour, and remove the last pattern once it is empty. With SONG on a chip wears its region's name and colour, and renaming or colouring it changes the region. **ON** switches the sequencer on and off; the pattern on screen is the one playing, unless the regions or a key decide.

Switching patterns from a keyboard. KEYS sets a range of sixteen notes — from C-1, C0, G#0, E7 or E8 — that pick a pattern: at the next step (*Change the pattern*), with step 1 at the switch (*Change + restart*) or at the next beat. G#0 ends just below a kit's pads and E7 starts just above bank D, so a full four-bank kit keeps every pad; a range that a pad sits on says so in the menu, because a trigger note never reaches the pads. While KEYS is on, a row of sixteen keys runs across the panel under the pattern chips, one per pattern, white and black where the notes fall, each wearing its pattern's colour and number: the pattern playing is lit, a pressed key goes amber while the right end shows which pattern is next, and a click on a key puts its pattern on screen. The outgoing pattern is cut at the switch.

Presets. The strip on the second row steps through the factory patterns — the arrows, or the wheel over the name — and a click on the name opens the list inside the panel, with the role chips Main, Fill, Build, Break and Perc and a typed filter (just type; Backspace; Escape closes). A preset lands in the pattern on screen by drum: its kick row goes to the lane aimed at the kick, its hats to the hats, wherever the kit on screen put them, in whichever bank is on screen; its swing becomes the pattern's SWING. Needs the Kit Bridge.

Regions and the song. SONG hands the choice of pattern to the regions: region 1 plays pattern 1, region 2 pattern 2, in time order; a gap plays nothing; a region that starts off the bar still starts its pattern on step

1. The song strip appears under the toolbar as a small timeline: each region drawn to its length with its pattern's number and its name, the playing region ringed and the playhead running across. Click a region to see its pattern, right-click for *Play from here*, *Add to the chain* (once, twice or four times), *Remove from the chain*, *Zoom to it*, *Rename* and *Delete*. The **+** adds a region after the last, named after its pattern, and the chain's own play, stop, loop and clear sit at the strip's left, under CHAIN. Needs the Kit Bridge; the Drum Matrix reads the same regions.



The piano roll. Right-click a lane's name and pick *Piano roll: notes, chords, lengths*: the lane folds out at the bottom of the panel as a roll with a keyboard, the other lanes shrinking above it and a grabber between them for the roll's height (drag; the wheel for small steps; double-click for half and half). Click a row to add a note. Drag a note to move it — **SNAP** decides where it lands: a step, a half, a third, or off — and drag its right third for its length; right-click deletes it. Up to six notes can start on one step: a chord, which a Repitch or Tuned pad plays as one; a pad with a key range plays one note at a time. **OCT** moves the window an octave, **SCALE** opens a menu of nine scales and a root — rows off the scale dim and a moved note snaps to it — and **FOLD** closes the roll. The lane's step grid is kept underneath; the same menu row turns the roll off again.

A lane as MIDI out. *MIDI out on a channel…* in the lane's menu makes the lane send its notes out of Swing on that channel instead of playing a pad — a bass line to the synth on the next track, fed from Swing's MIDI output. Note lengths hold, and stopping ends every note.

Saving. Patterns, the song chain, mute and solo, the options and the view are part of the project. **SAVE KIT** writes the patterns and the chain into the kit file too, so a kit comes back with its grooves; a kit saved before 3.0.19 leaves the grid alone.

TO ITEM. The pattern on screen as a MIDI item on Swing's track, from the bar the edit cursor is in: one cycle of the longest lane, as it would play there. Right-click for two, four or eight cycles. With SONG on it writes the song instead: one item under every region, named after the region and holding the pattern the region plays; a region whose pattern has no hits gets none; right-click for the chain in play order, repeats included, laid out from the edit cursor. One undo point either way. The item's piano roll shows the pad names. Switch ON off to hear the items alone. Needs the Kit Bridge.

MIDI out of the sequencer. *MIDI out* in the OPT menu, on by default, sends every hit the sequencer plays as a note on channel 10 as well, its note-off at the gate — REAPER can record it, or a send can drive a second instrument. The row below it picks the channel, and the panel's header shows *MIDI out 10* with a lamp that blinks for every note that leaves. A note that comes straight back on that channel is dropped, so a send that loops into Swing never doubles a hit.

Transport. With the option on, the toolbar's right end carries REAPER's transport: to the start, play, stop, loop and a bar : beat readout; Space plays and stops. Through the Kit Bridge.

OPT. *Lane icons*; *Follow the playhead* (the page turns with the playhead; a press on the page arrows switches it off); *Transport buttons*; *MIDI out* and *MIDI out channel*; *Aim this pattern at bank A / B / C / D* (every lane keeps its pad number, the bank changes); *Re-aim the lanes when a kit load moves the pads*; *Lanes bottom-up, like the pads*; and the two placements, *Sequencer at the side of the pads* and *Browser below the pads*.

Where the panels sit. By default the sequencer opens under the pads and the browser and the pad panel at the side. The three placement options in OPT move them: the sequencer into the side panel (its toolbar and footer fold for the narrower column; with the editor or the browser open too they sit beside each other when the window has the room, as tabs until it does), the browser below the pads, and the pad editor below the pads, where the strip's chevron, drag handle and a chip per panel serve them as they serve the sequencer; when two panels are placed below, the one opened last is the one shown and the chips bring the other forward. The window grows for whichever side a panel takes, and the choices are saved with the project. With both on screen the keyboard follows the last click: a click on a step gives the arrows and Enter to the sequencer, a click in the browser gives them to its list.

Keys. In a floating or chain window: the arrows move the selection, Enter toggles the step, Delete clears it, `+` and `-` change its velocity, PgUp and PgDn turn the page, `s v p r n a` pick a view, `1`-`8` a pattern, Ctrl+C and Ctrl+V copy and paste, Space plays and stops. Space is REAPER's whenever the panel does not have it.

Undo. Every edit — a step, an op, a preset, a kit's patterns — is a REAPER undo point, through the Kit Bridge.

Without the Kit Bridge the grid, the patterns, the roll, MIDI out and the keys all work. Presets, SONG, TO ITEM, the transport row and the undo points need it, and their buttons say so.

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17. The Companion Plugins

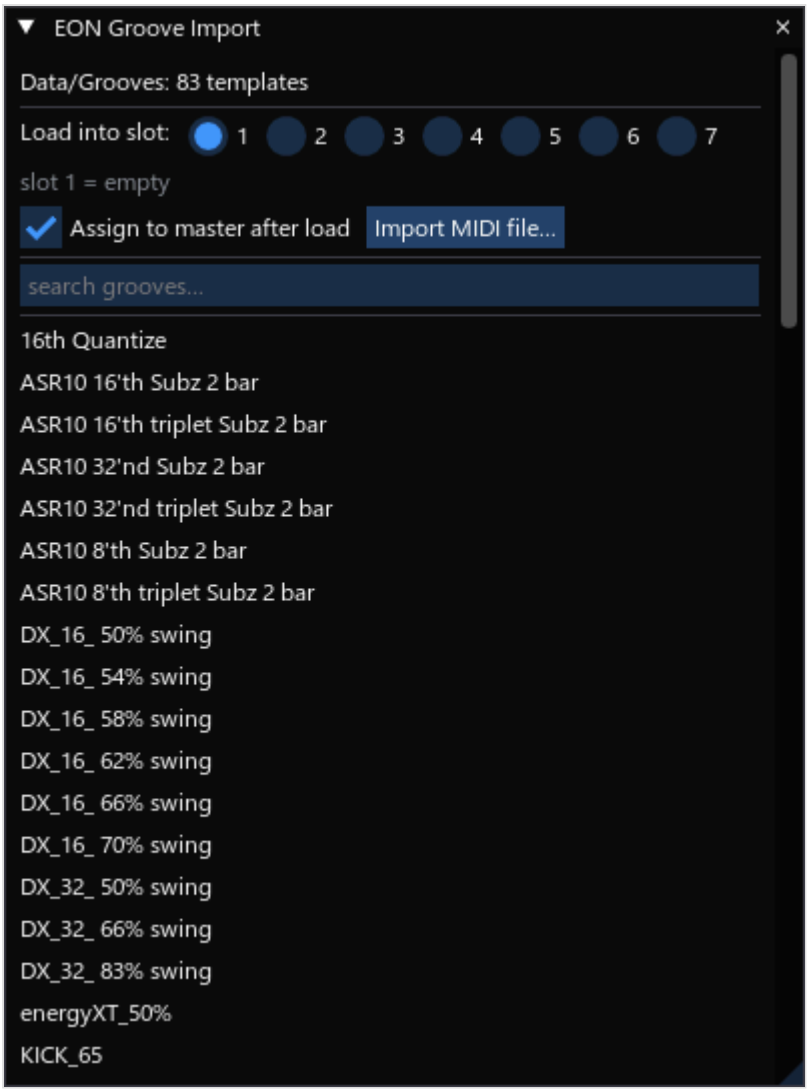
All of them are new in v3. Version 2.1 installed Swing and nothing else; since 3.0.19 Steppa installs on its own.

Steppa (free)

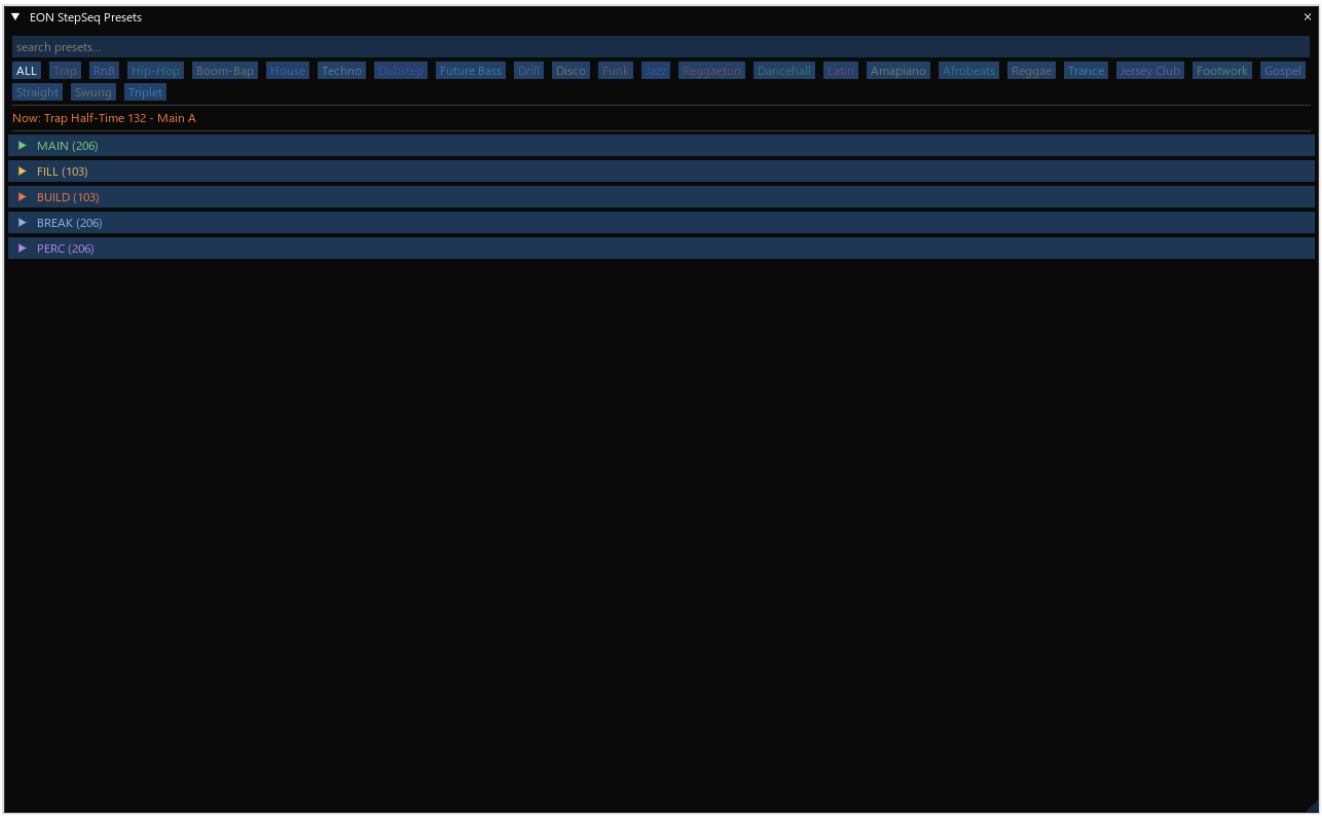


EON Steppa, the step sequencer that shipped inside Swing 3.0, is a free plugin of its own since 3.0.19 and no longer part of the Swing installer. Swing's own sequencer (chapter 16) covers most of what Steppa did for a kit and keeps the pads in one window; Steppa keeps the things that stayed outside it for now, and still pairs with Swing: the lanes take the pad names, colours and icons from the kit, pad edits mirror both ways, and the Drum Matrix reads its patterns. A project that has Steppa on a track opens as before — the free download writes the same file to the same place.

Groove system. A swing and groove engine with `.rgt` import *and* export, plus groove import straight from a MIDI file.



Preset browser — 824 factory patterns organised by role — main, fill, build, break, perc — with genre chips and live counts, in a ReaImGui window.



Song mode. Named arrangements that switch at the play-next boundary.

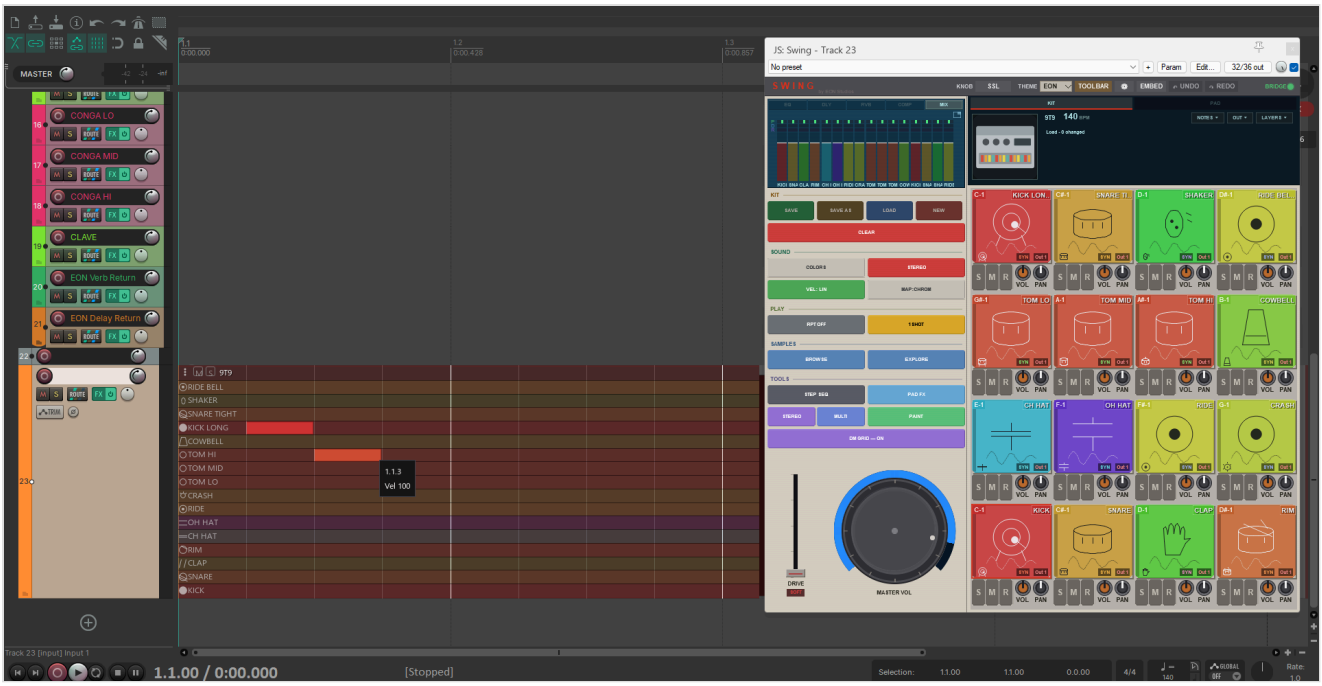
Controller lanes and user presets — the two things Swing's own sequencer does not do yet.

Drum Matrix

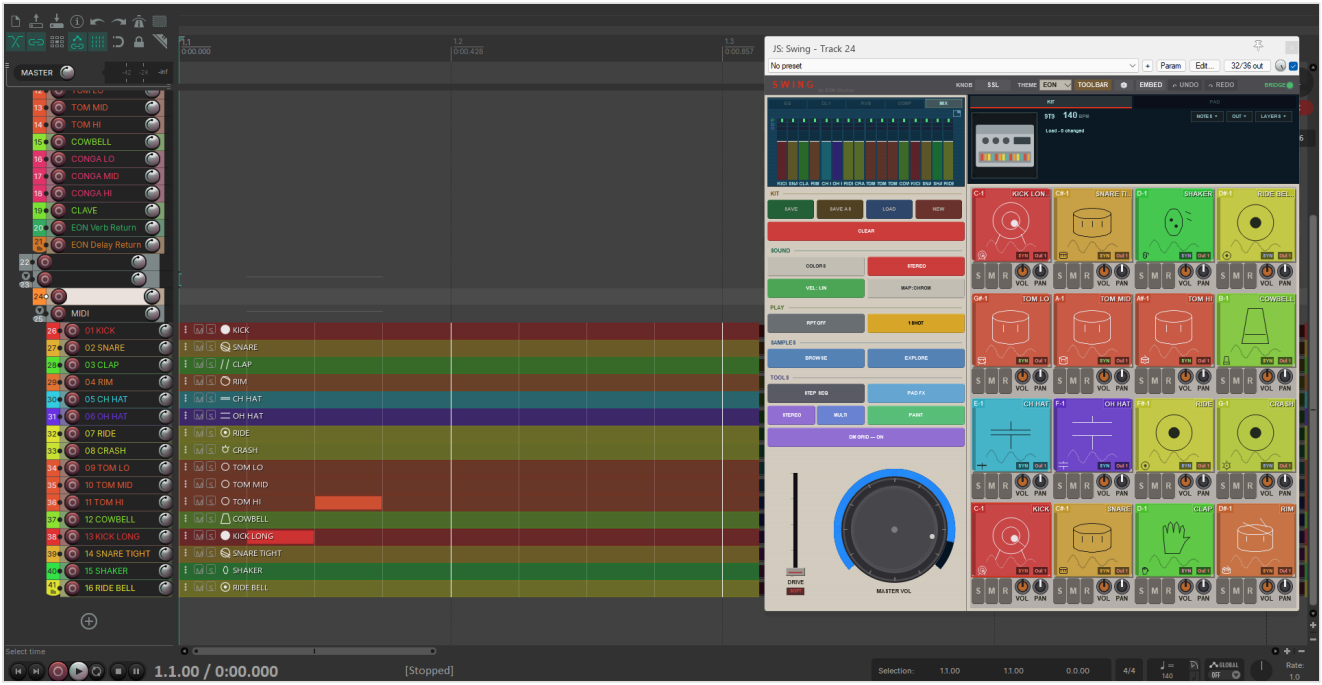
Your patterns as a grid in the arrange window, working on real REAPER MIDI items rather than a private format. Lanes carry your pad names, colours and icons straight through from Swing. Edit either end — sequencer or matrix — and the other follows, ratchets included. Role-based remapping means a pattern aimed at one kit lands sensibly on another.

Two ways to lay it out

Stereo grid keeps everything on the Swing track: one MIDI item, with a lane per pad inside it. Nothing else is added to your project. Build it from the tools row with **STEREO**, or right-click → **Build Stereo Grid (this track)**.



MIDI lanes gives each pad its own REAPER track — sixteen of them, named and coloured from the kit. Build it with **MULTI**, or right-click → **Build Drum Matrix (MIDI lanes)**.

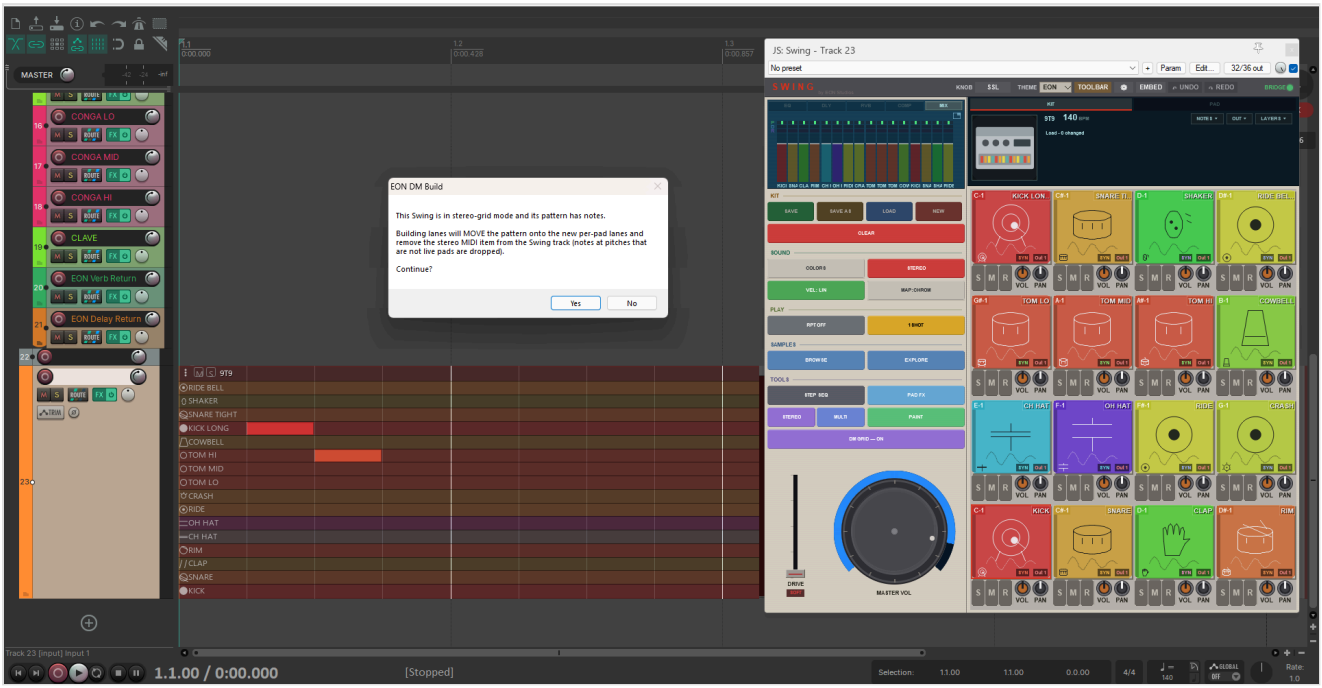


The right-click menu on those buttons also offers **Build Multi-Out (Audio)** and **Build Both (Audio + MIDI)**.

Banks. The Drum Matrix lanes are bank A's. While Swing shows another bank the two-way sync pauses and the builds refuse with a notice; show bank A and they carry on. One lane per pad across all four banks is a later step.

Switching from stereo grid to lanes

If the Swing is already in stereo-grid mode *and* its pattern has notes, Swing stops and asks before going any further, because the change is not additive:



> Building lanes will **move** the pattern onto the new per-pad lanes and remove

> the stereo MIDI item from the Swing track. **Notes at pitches that are not live

> pads are dropped.**

So a pattern written against a kit with empty pads, or against a different kit, can lose notes in the move. Answer No and nothing is touched.

When the build finishes it tells you how many lanes are ready and asks you to **load or restart** `eon_drum_matrix.lua` to display them — the lanes exist in the project either way, but the Drum Matrix window has to be re-opened to pick them up.

Drum Strip



The per-pad chain as its own plugin, and a full embedded track and mixer view. SSL-style knobs, EQ graph, gain-reduction meter, HPF/LPF nodes and VU-face selectors. One strip can broadcast its layout to the others, so a whole kit's strips match in one move.

FX Return View

Hardware-faceplate return monitors: the **EON 480** reverb deck and the **EON H9** delay deck, each showing every contributing pad's send on its own fader. Drop it on a return track and you see, at a glance, which pads are feeding that return and how hard — and you can ride the sends from there.

EON Lens

A kit-artwork card for the track panel. Your kit's cover, living on the track, pulsing with the audio. A kit without a cover gets a generated monogram, so the card never sits blank.

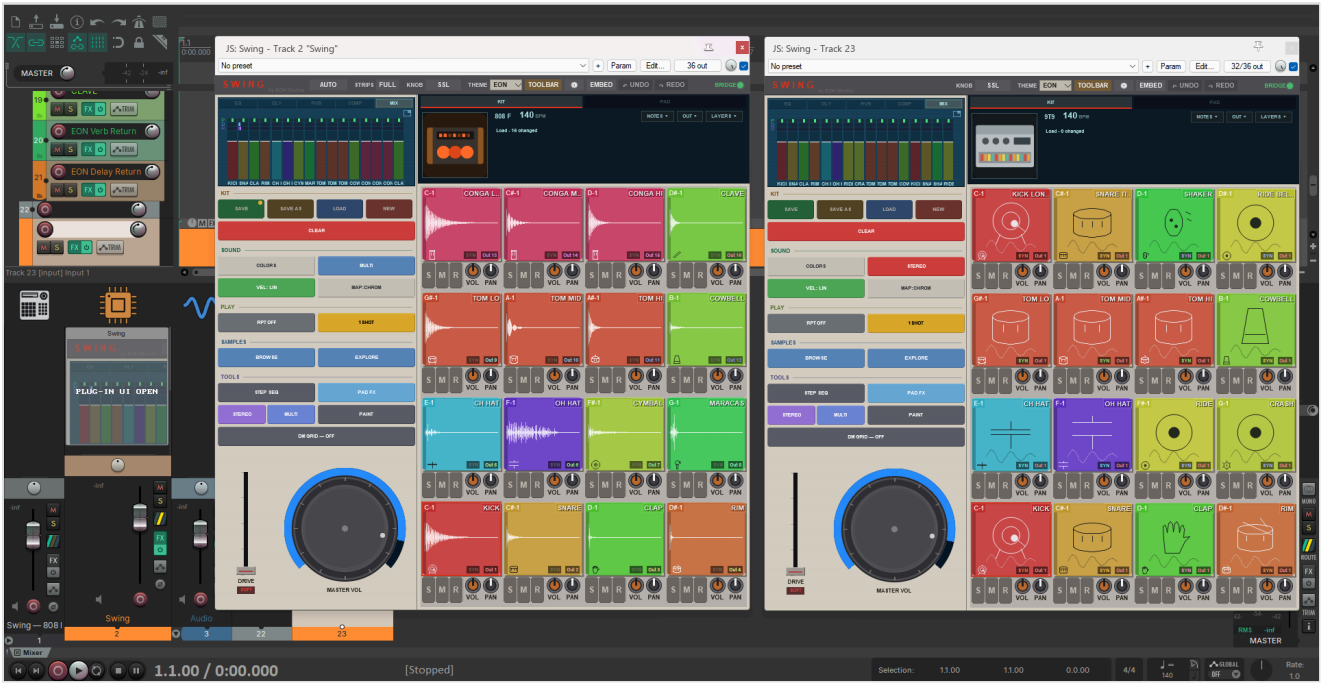
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18. Multi-Instance & Undo

New in v3.

Several Swings in one project

Every instance is its own instrument. Each keeps its own four banks, its own pitch state, mute and solo, colours and note names, and syncs two-way with the Drum Matrix independently. Nothing leaks between them.



Here the same project runs Swing on track 2 with the Fischer 808 kit and on track 23 with the 9T9 synth kit — different pads, different notes, different output routing, no interference.

The Sample Browser targets one instance at a time, so a dropped sample lands where you meant it to.

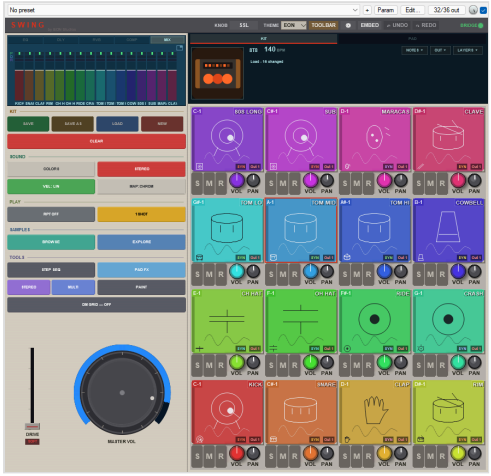
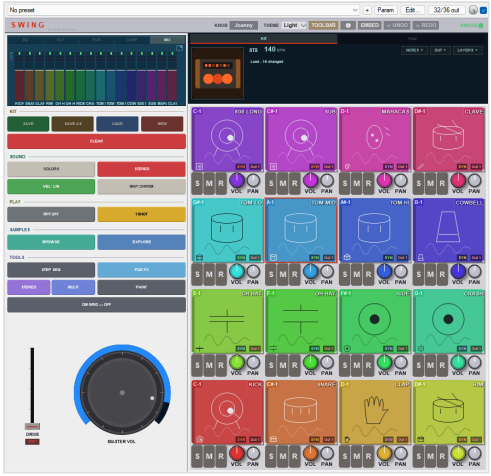
Undo

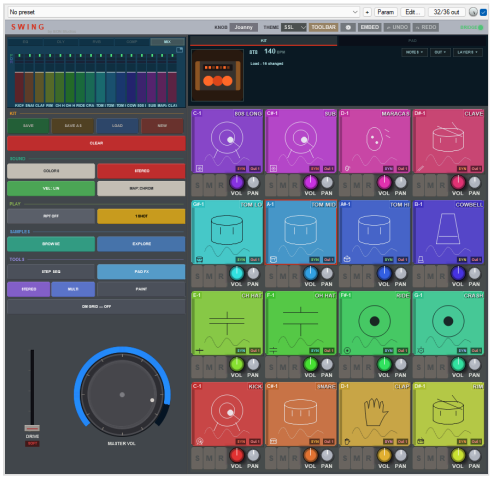
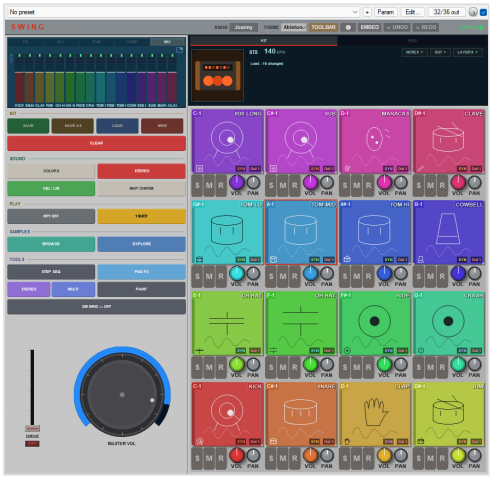
Swing has its own undo engine — parameter, structural and full-kit — pulled out of REAPER's global undo. Ctrl+Z can no longer wipe a kit.

19. Themes, Knobs & Controllers

Themes

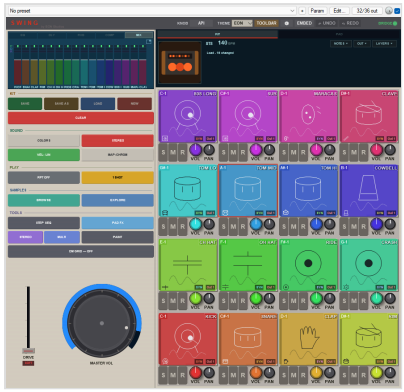
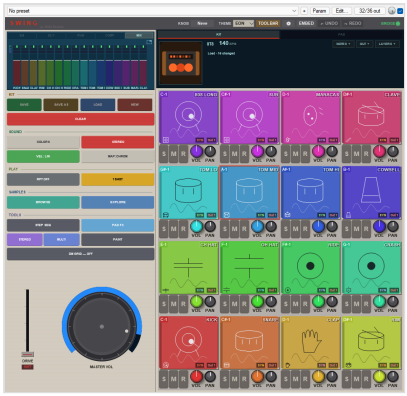
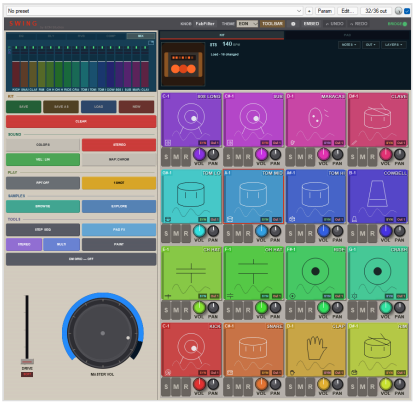
Fourteen console themes, each shipping its own knob style.

EON	Light
	

SSL	Ableton
	

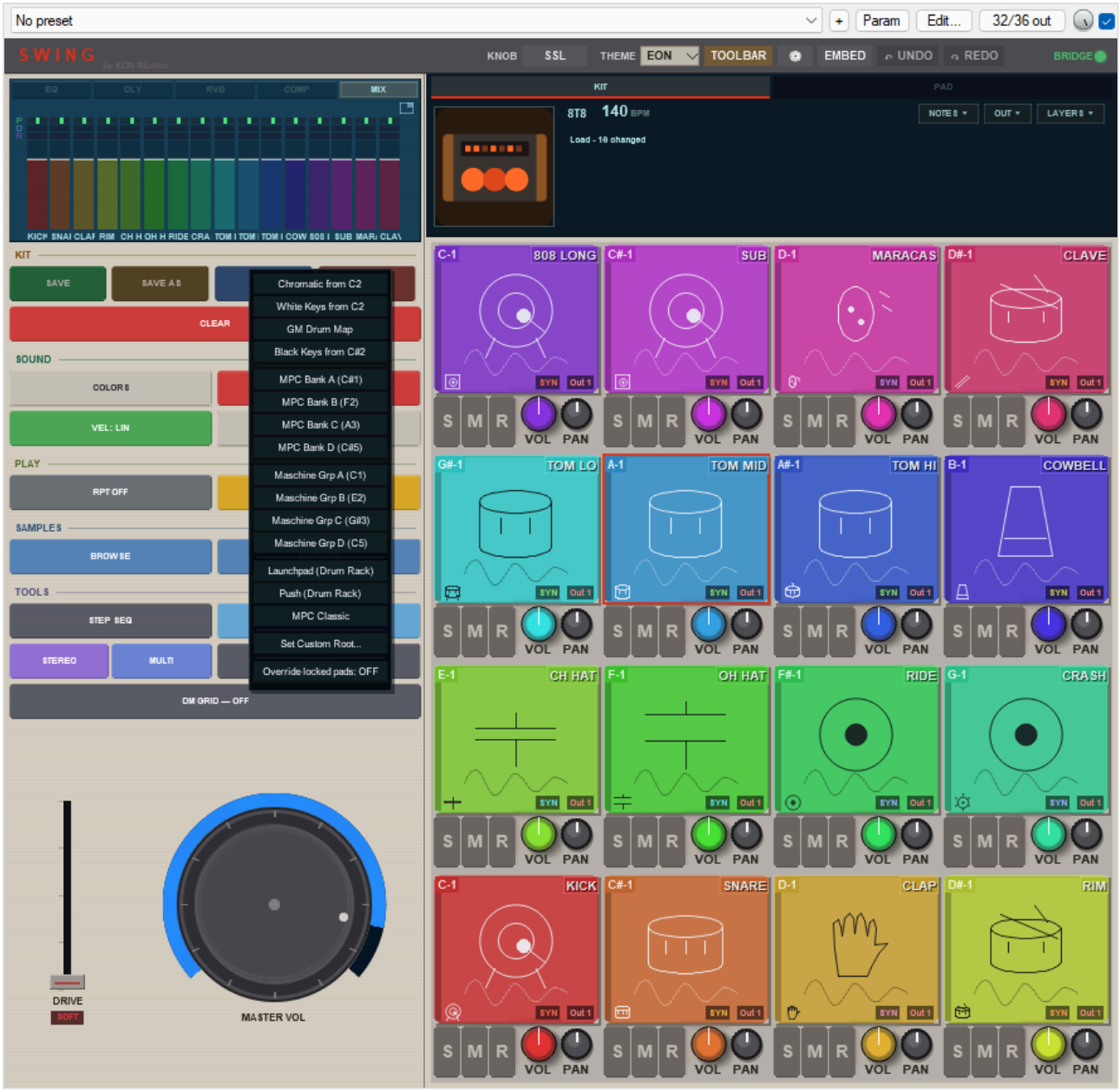
Knob styles

Knob style is separately selectable from the theme — eighteen of them.

API	Neve	FabFilter
		

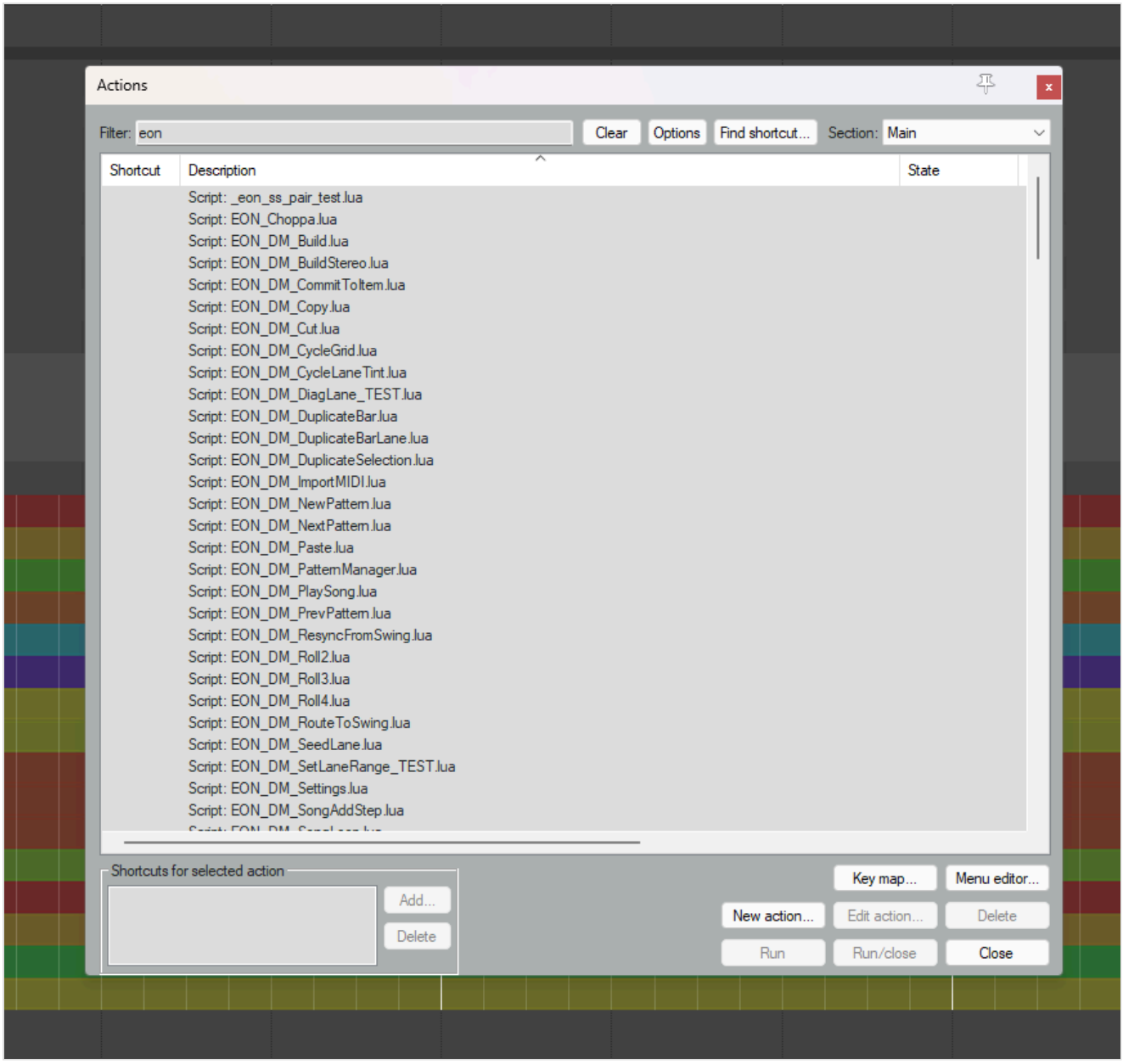
Per-pad knob colour. Right-click a pad's VOL or PAN knob — left half is VOL, right half is PAN — for that knob's own style and colour. **Follow Pad** tints the knob with the pad's hue. The two knobs are independent and the choice persists.

Controller presets



Sixteen presets covering MPC, Maschine, Launchpad and Push, plus a custom root note and an override-locked-pads toggle. A MIDI-learn mode steps through the pads so you can map a controller in one pass.

Actions and the toolbar



The whole catalogue is keybindable and toolbar-ready; the Kit Bridge registers it on its first run. New in v3: **New Song** (lays out tempo, four named regions, a Swing track, a MIDI item per region and the drum lanes in one action), **Install / Refresh Toolbar**, and the eleven view actions — one per full-window view, plus one that closes whichever is open.

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20. Recipes

Starting points, not rules.

Tight kick. Offset forward to remove silence at the front; HPF 25–35 Hz; a touch of Tape drive; keep decay short if the sample is long.

Punchy snare. Mid EQ +2–4 dB around 1–3 kHz; Soft or Tube drive in small amounts; short room reverb send.

Lo-fi hats. Bit depth 8–12 bits; gentle LPF to tame aliasing; small reverb send.

808 sub. Minimal drive; low shelf EQ +1–2 dB; long release; route to its own output to manage the low end.

Wide clap. Slight pan spread across layers in Sum mode; reverb width up; pre-delay for separation.

A synth kick from scratch. Switch the pad to synth and pick the kick family. Start with tune and decay — those two get you most of the way. Add drive last; it is easy to overdo when there is no sample noise to hide behind.

Hybrid pad. Layer a synth voice under a sample for weight the sample lacks. Set the synth low and let the sample keep the transient.

A playable 808. Give the pad a key range, leave it in Repitch, and play the line — one note at a time, since ranged pads are monophonic.

Hat pair. Put the open and closed hats in the same choke group. That is the whole trick.

Parallel crush. Send a pad into a return and crush the return rather than the pad, so the clean hit stays underneath.

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21. Troubleshooting

Plugin not found. Restart REAPER. Verify the files exist — under `Effects/EON/Swing` for an installer setup, or `Effects/EON Studios – Swing 2/EON/Swing` for a ReaPack install — and re-scan if needed.

Browser won't open. Requires ReaImGui — install via ReaPack.

Browser opens but preview is silent. That is SWS missing — install it via ReaPack and audition works.

Drum Matrix won't open. Requires js_ReaScriptAPI, via ReaPack.

Kit Bridge doesn't start. The Windows installer registers it to start with REAPER; on a ReaPack install it starts after you run `Swing_Kit_Bridge` once from the Action List. Run it once manually and it registers itself from then on.

Stretch sounds like Repitch. The EON engine is not installed. Tuned and Stretch fall back to Repitch without it, and Swing shows a notice saying so.

Old kits sound different. Swing loads 2.1-era kits with neutral defaults for anything that did not exist then — filters wide open, EQ flat, saturation off, no sends.

A pad plays only one note of a chord. Expected. Ranged pads are monophonic.

Sound cuts off when you switch to another application. That one is REAPER, not Swing. **Options → Preferences → Audio → Device** has **Close audio device when stopped and application is inactive**, and it is on by default: with the transport stopped, REAPER hands the audio device back to the OS the moment it loses focus, which kills anything still sounding — a long sample you just auditioned, a reverb tail. Uncheck it. The trade-off is that REAPER keeps the device claimed for as long as it is open, so other apps may go silent instead. Playing sounds are unaffected while the transport is rolling, which is a quick way to confirm this is what you are seeing.

WIRED? shows in amber. The EON Patchbay is not answering: it must be last in the Swing track's FX chain and enabled. Until it is, pads sum inside Swing and the pad inserts are silent.

Own output made no track. The pad has no sound on it yet, or you deleted the track by hand — build again and choose Update — or you just undid something: the track is made after your next action, so your redo history is kept.

Waiting: all 16 Tuned / Stretch spots in use. Sixteen pads across the four banks already hold the pitch engine's spots. The pad plays Repitch until one is freed, then takes it.

Pads came back blank after reopening a project. Fixed in v3. If you see it on a 3.0 install, check that the Kit Bridge is running.
